

Mathematical Methods for Finance

Prerequisites Package — Extended Self-Study Edition

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Based on the undergraduate courses in General Mathematics and Statistics

Detailed reference with abundant visual aids, worked examples, diagnostic exercises and solutions

How this extended package is organized

Purpose

This is a reference package, not a list of topics that must be re-taught during the course. Each topic is presented in five steps: explanation, worked example, checks and common mistakes, exercises, and concise solutions.

- The mathematical part follows the undergraduate General Mathematics syllabus and adds a dedicated trigonometry chapter.
- The integration chapter includes substitution, integration by parts, rational functions, partial fractions, trigonometric methods, reduction formulae and improper integrals.
- The statistics part expands descriptive statistics, probability, random variables, inference and regression so students can use the package independently when a gap emerges.

Index of the Prerequisites

Orientation and method

Sets, numbers and topology of the real line

Functions and elementary functions

Trigonometry

Sequences, limits and continuity

Differential calculus and optimization

Integrals and integration techniques

Linear algebra

Functions of several variables

Descriptive statistics

Probability

Random variables and probability models

Sampling and statistical inference

Linear regression and statistical interpretation

Bridge to Mathematical Methods for Finance

Chapter 1

Orientation and method

1 detailed review units $\times$ five frames per unit

How to use the prerequisite package — overview

Concept

This package is a self-study reference. It is deliberately more detailed than the material that will be reviewed in class.

- Read the definition first; then reproduce the worked example without looking.
- Use the exercises diagnostically: a mistake identifies a topic to revise.
- Keep a personal formula sheet, but always record the assumptions next to each formula.

How to use the prerequisite package — example

Problem

A student remembers the derivative of $\log x$ but forgets that $x > 0$. What should be written on the formula sheet?

Worked solution

Write both pieces: $\frac{d}{dx} \log x = \frac{1}{x}$ *for* $x > 0$. The domain is part of the mathematical statement.

How to use the prerequisite package — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not memorize formulas without domains.
- Do not skip definitions because an example looks familiar.
- Separate computational fluency from conceptual understanding.

How to use the prerequisite package — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 List three prerequisites you expect to use in probability.
- 2 Give one example in which a domain restriction changes the answer.
- 3 Explain the difference between “I can follow a solution” and “I can solve the problem”.

How to use the prerequisite package — solutions

Concise solutions

- ① Typical answers: sets/events, functions/random variables, sums/integrals.
- ② For $\sqrt{x-2}$ the natural domain is $[2, \infty)$; values below 2 are inadmissible over $\mathbb{R}$.
- ③ Independent solution requires selecting the method and checking the result without a model solution.

Chapter 2

Sets, numbers and topology of the real line

10 detailed review units $\times$ five frames per unit

Concept

A set is a collection of distinct objects. Membership is written $x \in A$ and non-membership $x \notin A$. A set can be described by listing elements or by a defining property.

- Roster notation: $A = \{1, 2, 3\}$.
- Set-builder notation: $B = \{x \in \mathbb{R} : x^2 < 4\}$.
- The empty set $\emptyset$ has no elements.

Sets and membership — example

Problem

Describe $A = \{x \in \mathbb{Z} : -2 \leq x < 3\}$ by listing its elements.

Worked solution

The admissible integers are $-2, -1, 0, 1, 2$, hence $A = \{-2, -1, 0, 1, 2\}$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not confuse $x \in A$ with $\{x\} \subseteq A$.
- Repeated entries do not create new elements.
- Always state the ambient number system when inequalities are used.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Write $\{2, 4, 6, 8\}$ in set-builder notation.
- 2 List the elements of $\{x \in \mathbb{N} : x < 5\}$ under the convention $0 \in \mathbb{N}$.
- 3 Is $\emptyset$ finite or infinite?

Concise solutions

- ① $\{x \in \mathbb{N} : 2 \leq x \leq 8, x \text{ even}\}.$
- ② $\{0, 1, 2, 3, 4\}.$
- ③ Finite: it has cardinality 0.

Subsets and set equality — overview

Concept

$A \subseteq B$ means that every element of A belongs to B . Two sets are equal exactly when each is a subset of the other.

- $A = B \iff A \subseteq B$ and $B \subseteq A$.
- A proper subset satisfies $A \subseteq B$ and $A \neq B$.
- The power set $\mathcal{P}(A)$ is the set of all subsets of A .

Subsets and set equality — example

Problem

Let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$. Decide whether $A \subseteq B$ and compute $\mathcal{P}(A)$.

Worked solution

Yes. Moreover $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Membership and inclusion use different symbols.
- $\emptyset \subseteq A$ for every set A .
- $A \subseteq A$ is always true; proper inclusion is different.

Subsets and set equality — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If A has 4 elements, how many subsets does it have?
- 2 Prove that $\{1, 2\} = \{2, 1\}$.
- 3 Is $\{1\} \in \mathcal{P}(\{1, 2\})$?

Concise solutions

- ① $2^4 = 16$.
- ② Both sets have exactly the same elements; order is irrelevant.
- ③ Yes, because $\{1\} \subseteq \{1, 2\}$.

Union, intersection, difference and complement — overview

Concept

Set operations combine logical conditions. Union corresponds to “or”, intersection to “and”, and complement to “not” relative to a chosen universe.

- $A \cup B = \{x : x \in A \text{ or } x \in B\}$.
- $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
- $A \setminus B = \{x \in A : x \notin B\}$.
- De Morgan: $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Union, intersection, difference and complement — example

Problem

For $A = [0, 3]$ and $B = (2, 5)$, compute $A \cap B$ and $A \cup B$.

Worked solution

$A \cap B = (2, 3]$ and $A \cup B = [0, 5)$.

Union, intersection, difference and complement — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Endpoint inclusion must be checked one endpoint at a time.
- A complement is meaningless unless the universe is understood.
- $A \setminus B$ is not the same as $B \setminus A$.

Union, intersection, difference and complement — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $(-1, 2] \cap [2, 4)$.
- 2 Compute $\mathbb{R} \setminus (-\infty, 3]$.
- 3 Verify one De Morgan identity for $A = \{1, 2\}$, $B = \{2, 3\}$ in $U = \{1, 2, 3, 4\}$.

Union, intersection, difference and complement — solutions

Concise solutions

- ① $\{2\}$.
- ② $(3, \infty)$.
- ③ $(A \cup B)^c = \{4\} = A^c \cap B^c$.

Cartesian products and ordered pairs — overview

Concept

The Cartesian product $A \times B$ contains ordered pairs (a, b) with $a \in A$ and $b \in B$. Order matters.

- $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$.
- $|A \times B| = |A||B|$ for finite sets.
- Graphs of functions are subsets of Cartesian products.

Cartesian products and ordered pairs — example

Problem

If $A = \{1, 2\}$ and $B = \{a, b, c\}$, list $A \times B$.

Worked solution

$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- (a, b) and (b, a) are generally different.
- $A \times B$ and $B \times A$ need not be equal.
- Do not replace ordered pairs by two-element sets.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 How many elements does a 5-element set times a 7-element set have?
- 2 Describe $[0, 1] \times [0, 1]$ geometrically.
- 3 Is $(1, 2) \in \{1, 2\} \times \{1, 2\}$?

Concise solutions

- ① 35.
- ② The closed unit square in the plane.
- ③ Yes.

Natural, integer, rational, real and complex numbers — overview

Concept

The standard number systems form the chain $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. Each extension permits operations or equations that may fail in the preceding system.

- Rationals have the form p/q with integers p, q and $q \neq 0$.
- Irrational real numbers cannot be represented as such a ratio.
- Complex numbers have the form $a + ib$ with $i^2 = -1$.

Natural, integer, rational, real and complex numbers — example

Problem

Classify -3 , $2/7$, $\sqrt{2}$, and $3 - 4i$ in the smallest standard set containing each.

Worked solution

$-3 \in \mathbb{Z}$, $2/7 \in \mathbb{Q}$, $\sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$, and $3 - 4i \in \mathbb{C} \setminus \mathbb{R}$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not call every decimal irrational: terminating and repeating decimals are rational.
- The convention for whether $0 \in \mathbb{N}$ varies.
- A real number is also a complex number with zero imaginary part.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Classify 0.125.
- 2 Classify π .
- 3 Solve $x^2 + 1 = 0$ in $\mathbb{R}$ and in $\mathbb{C}$.

Concise solutions

- ① $0.125 = 1/8 \in \mathbb{Q}$.
- ② π is irrational, hence real.
- ③ No real solutions; in $\mathbb{C}$, $x = \pm i$.

Order, intervals and absolute value — overview

Concept

The real numbers are ordered. Intervals encode inequalities, while $|x|$ measures the distance from x to 0.

- $|x - a| < r \iff a - r < x < a + r.$
- $|x| \geq 0$ and $|xy| = |x||y|.$
- Triangle inequality: $|x + y| \leq |x| + |y|.$

Order, intervals and absolute value — example

Problem

Solve $|x - 2| \leq 3$.

Worked solution

The inequality means that the distance from x to 2 is at most 3: $-3 \leq x - 2 \leq 3$, hence $x \in [-1, 5]$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not square an inequality before checking whether it preserves equivalence.
- Open and closed endpoints are part of the answer.
- Absolute value inequalities are distance statements.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Solve $|2x + 1| < 5$.
- 2 Write $(-2, 4]$ as inequalities.
- 3 Find all x such that $|x| > 2$.

Order, intervals and absolute value — solutions

Concise solutions

- ① $-3 < x < 2$.
- ② $-2 < x \leq 4$.
- ③ $(-\infty, -2) \cup (2, \infty)$.

Bounds, supremum, infimum, maximum and minimum — overview

Concept

An upper bound lies above every point of a set. The supremum is the least upper bound. A maximum is a supremum that actually belongs to the set; analogous definitions hold below.

- For $(0, 1)$, $\sup A = 1$ but $\max A$ does not exist.
- For $[0, 1]$, $\sup A = \max A = 1$.
- Completeness of $\mathbb{R}$ guarantees the existence of suprema for nonempty bounded-above sets.

Bounds, supremum, infimum, maximum and minimum — example

Problem

Find inf, sup, minimum and maximum of $A = \{1/n : n \in \mathbb{N}, n \geq 1\}$.

Worked solution

$\sup A = \max A = 1$. The values approach 0 but never reach it, so $\inf A = 0$ and no minimum exists.

Bounds, supremum, infimum, maximum and minimum — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Supremum and maximum are not synonyms.
- A set can be bounded without having extrema.
- Always check whether the candidate bound belongs to the set.

Bounds, supremum, infimum, maximum and minimum — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find extrema and bounds for $(2, 7)$.
- 2 Find $\sup\{x \in \mathbb{R} : x^2 < 4\}$.
- 3 Does an unbounded-above set have a real supremum?

Bounds, supremum, infimum, maximum and minimum — solutions

Concise solutions

- ① $\inf = 2$, $\sup = 7$, no minimum or maximum.
- ② 2.
- ③ No; one sometimes writes $+\infty$ in the extended real line, but it is not a real number.

Neighborhoods, interior, boundary and accumulation points — overview

Concept

Local properties on the real line are described with neighborhoods. For $\varepsilon > 0$, $I_\varepsilon(x_0) = (x_0 - \varepsilon, x_0 + \varepsilon)$.

- Interior point: some neighborhood lies entirely in the set.
- Boundary point: every neighborhood meets the set and its complement.
- Accumulation point: every neighborhood contains a different point of the set.
- An isolated point belongs to the set but is not an accumulation point.

Neighborhoods, interior, boundary and accumulation points — example

Problem

For $A = (0, 1) \cup \{2\}$, identify interior, boundary, accumulation and isolated points.

Worked solution

$\text{int } A = (0, 1)$. Boundary points are $\{0, 1, 2\}$. Accumulation points include every point of $[0, 1]$; 2 is isolated.

Neighborhoods, interior, boundary and accumulation points — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- An accumulation point need not belong to the set.
- An isolated point can still be a boundary point.
- Interior and boundary depend on the ambient space.

Neighborhoods, interior, boundary and accumulation points — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the accumulation points of $\{1/n : n \geq 1\}$.
- 2 Is 0 an interior point of $[0, 1]$ in $\mathbb{R}$?
- 3 Identify the isolated points of $\mathbb{Z}$.

Neighborhoods, interior, boundary and accumulation points — solutions

Concise solutions

- ① Only 0.
- ② No; every neighborhood of 0 contains negative numbers.
- ③ Every integer is isolated.

Concept

A subset is dense in $\mathbb{R}$ if every nonempty open interval contains one of its points. Countability concerns whether elements can be indexed by natural numbers.

- $\mathbb{Q}$ is countable and dense in $\mathbb{R}$.
- The irrational numbers are also dense in $\mathbb{R}$.
- $\mathbb{R}$ is uncountable: no sequence can list all real numbers.

Dense, countable and uncountable sets — example

Problem

Can a countable set be dense? Explain using $\mathbb{Q}$.

Worked solution

Yes. Density concerns how points are distributed, not how many there are in the cardinality sense. The rationals are countable yet every open interval contains rational numbers.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not confuse “dense” with “uncountable”.
- An infinite set need not be uncountable.
- Cardinality and geometric spacing are different concepts.

Dense, countable and uncountable sets — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Is $\mathbb{Z}$ dense in $\mathbb{R}$?
- 2 Is every finite set countable?
- 3 Give a dense subset of $(0, 1)$.

Concise solutions

- ① No; for example $(0.1, 0.2)$ contains no integer.
- ② Yes.
- ③ $\mathbb{Q} \cap (0, 1)$ is one example.

Complex numbers and the fundamental theorem of algebra — overview

Concept

For $z = a + ib$, the conjugate is $\bar{z} = a - ib$ and the modulus is $|z| = \sqrt{a^2 + b^2}$. Complex numbers close polynomial root problems.

- $z\bar{z} = |z|^2$.
- Division is performed by multiplying numerator and denominator by a conjugate.
- Every nonconstant polynomial with complex coefficients has a complex root; degree n gives n roots counting multiplicity.

Complex numbers and the fundamental theorem of algebra — example

Problem

Compute $(2 + 3i)/(1 - i)$.

Worked solution

Multiply by $1 + i$: $\frac{(2+3i)(1+i)}{(1-i)(1+i)} = \frac{-1+5i}{2} = -\frac{1}{2} + \frac{5}{2}i$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Remember $i^2 = -1$.
- Complex conjugation changes the sign of the imaginary part only.
- Multiplicity matters when counting polynomial roots.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $|3 - 4i|$.
- 2 Solve $z^2 - 4z + 13 = 0$.
- 3 Compute $(1 + i)^2$.

Complex numbers and the fundamental theorem of algebra — solutions

Concise solutions

- ① 5.
- ② $z = 2 \pm 3i$.
- ③ $2i$.

Chapter 3

Functions and elementary functions

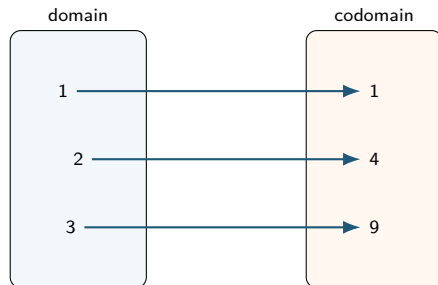
9 detailed review units $\times$ five frames per unit

Functions, domain, codomain and image — overview

Concept

A function assigns to each element of the domain exactly one image in the codomain.

- The domain is the set of admissible inputs.
- The codomain is the declared target set.
- The image is the subset of outputs actually attained.



Example: $f(x) = x^2$ on $\{1, 2, 3\}$

Functions, domain, codomain and image — example

Problem

Find the natural domain and image of $f(x) = \sqrt{4 - x^2}$.

Worked solution

The radicand requires $4 - x^2 \geq 0$, hence $x \in [-2, 2]$. Since $f(x) \geq 0$ and reaches 2 at $x = 0$, the image is $[0, 2]$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Never perform algebra before identifying domain restrictions.
- A formula alone does not completely specify a function unless its domain is understood.
- Image is an output set; domain is an input set.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the domain of $1/(x^2 - 9)$.
- 2 Find the domain of $\log(2 - x)$.
- 3 Find the image of x^2 on $[-1, 3]$.

Concise solutions

- ① $\mathbb{R} \setminus \{-3, 3\}$.
- ② $(-\infty, 2)$.
- ③ $[0, 9]$.

Concept

Solving an equation means identifying all admissible inputs for which two expressions agree. Solving an inequality means describing all admissible inputs satisfying an order relation.

- Factorization often turns a nonlinear equation into simpler factors.
- Sign charts organize rational and polynomial inequalities.
- Domain restrictions are imposed before simplification.

Equations and inequalities — example

Problem

Solve $\frac{x-1}{x+2} \geq 0$.

Worked solution

Critical points are $x = 1$ and $x = -2$; $x = -2$ is excluded. A sign chart gives $(-\infty, -2) \cup [1, \infty)$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not cancel a factor without preserving excluded values.
- Multiplying an inequality by an expression of unknown sign can reverse the inequality.
- Zeros and poles play different roles in sign charts.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Solve $(x - 2)(x + 1) < 0$.
- 2 Solve $x^2 - 5x + 6 = 0$.
- 3 Solve $1/(x - 1) > 2$.

Concise solutions

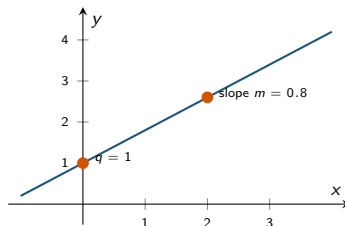
- ① $(-1, 2)$.
- ② $x = 2, 3$.
- ③ After bringing to one fraction: $(3 - 2x)/(x - 1) > 0$, hence $(1, 3/2)$.

Straight lines and affine functions — overview

Concept

A straight line has equation $y = mx + q$, where m is the slope and q is the intercept.

- $m > 0$: increasing line; $m < 0$: decreasing line.
- $|m|$ measures steepness: larger absolute slope means a steeper line.
- q is the value of y when $x = 0$.



Straight lines and affine functions — example

Problem

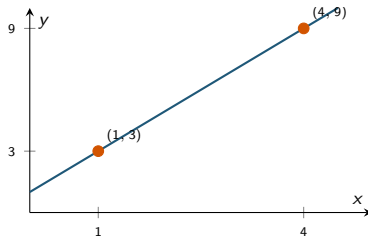
Find the line through $(1, 3)$ and $(4, 9)$.

Worked solution

The slope is $m = (9 - 3)/(4 - 1) = 2$. Using point-slope form,

$$y - 3 = 2(x - 1),$$

so the line is $y = 2x + 1$.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

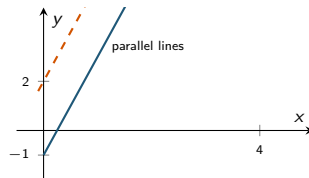
- A vertical line has equation $x = c$ and no finite slope.
- Do not confuse intercept with the value of x where the line crosses the x -axis.
- Units matter: a slope is output units per input unit.

Straight lines and affine functions — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the slope of $3x - 2y = 6$.
- 2 Find the line through $(0, -1)$ parallel to $y = 4x + 2$.
- 3 Find the x -intercept of $y = 2x - 8$.



Straight lines and affine functions — solutions

Concise solutions

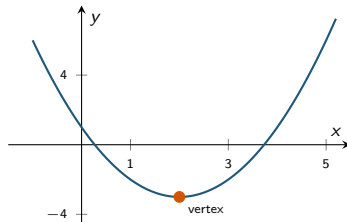
- ① $m = 3/2$.
- ② $y = 4x - 1$.
- ③ $x = 4$.

Quadratic functions, parabolas and completing the square — overview

Concept

A quadratic function has the form $f(x) = ax^2 + bx + c$ with $a \neq 0$. Its graph is a parabola.

- If $a > 0$, the parabola opens upward; if $a < 0$, downward.
- Completing the square gives vertex form $a(x - h)^2 + k$.
- The vertex identifies the turning point of the graph.



Quadratic functions, parabolas and completing the square — example

Problem

Rewrite $x^2 - 6x + 5$ in vertex form and identify its zeros.

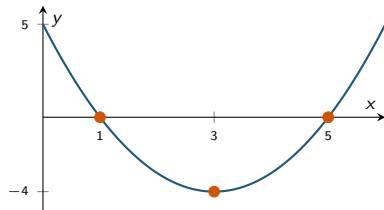
Worked solution

$$x^2 - 6x + 5 = (x - 3)^2 - 4.$$

Hence the vertex is $(3, -4)$. Factorization gives

$$x^2 - 6x + 5 = (x - 1)(x - 5),$$

so the zeros are 1 and 5.



Quadratic functions, parabolas and completing the square — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

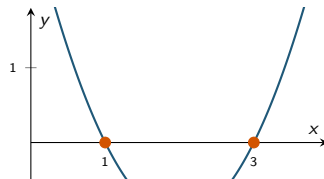
- The sign of a determines opening direction.
- Completing the square changes representation, not the function.
- A negative discriminant means no real roots, not no complex roots.

Quadratic functions, parabolas and completing the square — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Complete the square for $x^2 + 4x + 7$.
- 2 Find the vertex of $2x^2 - 8x + 1$.
- 3 For which x is $x^2 - 4x + 3 \leq 0$?



Quadratic functions, parabolas and completing the square — solutions

Concise solutions

- ① $(x + 2)^2 + 3$.
- ② $(2, -7)$.
- ③ $[1, 3]$.

Power, root and reciprocal functions — overview

Concept

Power functions x^α change their domains and qualitative behavior with the exponent. Roots and reciprocals are especially important because they introduce boundary and pole restrictions.

- $x^{-n} = 1/x^n$ for $x \neq 0$.
- Even roots require a nonnegative radicand over $\mathbb{R}$.
- Odd roots are defined for all real inputs.

Power, root and reciprocal functions — example

Problem

Determine the domain of $f(x) = \sqrt{x-1}/(x-3)$.

Worked solution

The root requires $x \geq 1$ and the denominator requires $x \neq 3$. Thus $D = [1, 3) \cup (3, \infty)$.

Power, root and reciprocal functions — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Fractional exponents can conceal root restrictions.
- A negative exponent introduces a denominator.
- Do not infer monotonicity without checking the domain.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the domain of $(x + 2)^{-1/2}$.
- 2 Simplify $x^{3/2}$ for $x \geq 0$.
- 3 Describe the asymptotes of $1/x$.

Concise solutions

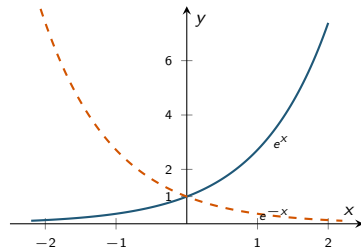
- ① $(-2, \infty)$.
- ② $x\sqrt{x}$.
- ③ Vertical $x = 0$ and horizontal $y = 0$.

Exponential functions — overview

Concept

For $a > 0$, $a \neq 1$, the exponential function a^x is positive on all of $\mathbb{R}$. It grows when $a > 1$ and decays when $0 < a < 1$.

- $a^{x+y} = a^x a^y$.
- e^x is the natural exponential.
- Continuous compounding produces factors such as e^{rt} .



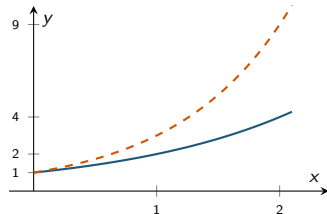
Exponential functions — example

Problem

Compare the values of 2^x and 3^x at $x = 0, 1, 2$ and explain which one grows faster.

Worked solution

At $x = 0$ both equal 1. At $x = 1$, $2^1 = 2$ and $3^1 = 3$. At $x = 2$, $2^2 = 4$ and $3^2 = 9$. Since the gap widens as x increases, 3^x grows faster than 2^x .



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- An exponential is never zero.
- Do not confuse e^{x+y} with $e^x + e^y$.
- Taking logs is valid only after confirming the relevant quantity is positive.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Solve $2^{x+1} = 16$.
- 2 Compare e^{-2} and e^{-1} .
- 3 Find t such that $100e^{0.05t} = 150$.

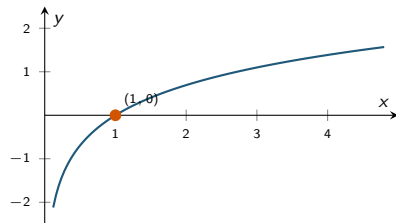
Concise solutions

- ① $x = 3$.
- ② $e^{-2} < e^{-1}$.
- ③ $t = \log(1.5)/0.05$.

Concept

The logarithm $\log_a x$ is the inverse of a^x . The natural logarithm $\log x$ has base e and domain $(0, \infty)$.

- $\log(xy) = \log x + \log y$ for $x, y > 0$.
- $\log(x/y) = \log x - \log y$.
- $\log(x^\alpha) = \alpha \log x$ when the expression is defined.



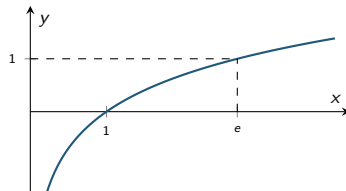
Logarithms — example

Problem

Solve $\log x = 1$ and interpret the answer graphically.

Worked solution

Since $\log x = 1$ means $x = e^1 = e$, the solution is $x = e \approx 2.718$. On the graph of $y = \log x$, this is the point where the curve reaches height 1.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- The logarithm of a sum does not split.
- Domain conditions can eliminate algebraic roots.
- Always distinguish $\log x^2$ from $(\log x)^2$.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Simplify $\log(e^{3x})$.
- 2 Solve $\log x = 2$.
- 3 State the domain of $\log(4 - x^2)$.

Concise solutions

- ① $3x$.
- ② $x = e^2$.
- ③ $(-2, 2)$.

Concept

The composition $(f \circ g)(x) = f(g(x))$ is defined where $g(x)$ belongs to the domain of f . An inverse function reverses a one-to-one map.

- Composition is generally not commutative.
- f^{-1} is not the reciprocal $1/f$.
- If f is strictly monotone on an interval, it is one-to-one there.

Composition and inverse functions — example

Problem

Let $f(x) = x^2$ and $g(x) = x + 1$. Compute $f \circ g$ and $g \circ f$.

Worked solution

$(f \circ g)(x) = (x + 1)^2$, while $(g \circ f)(x) = x^2 + 1$. They are different.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Check the domain of the outer function after substitution.
- An inverse may require restricting the original domain.
- Use $f(f^{-1}(y)) = y$ as a verification.

Composition and inverse functions — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the inverse of $f(x) = 3x - 2$.
- 2 Find an inverse for x^2 restricted to $[0, \infty)$.
- 3 Find the domain of $\sqrt{\log x}$.

Concise solutions

- ① $f^{-1}(y) = (y + 2)/3$.
- ② $f^{-1}(y) = \sqrt{y}, y \geq 0$.
- ③ $[1, \infty)$.

Concept

Monotonicity compares function values as inputs increase. Convexity compares the graph with its chords. These qualitative properties are central to optimization.

- Strictly increasing: $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$.
- Convexity: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$.
- The epigraph is $\text{epi}(f) = \{(x, t) : t \geq f(x)\}$.

Monotonicity, extrema, convexity and epigraphs — example

Problem

Show that $f(x) = x^2$ is convex by using its second derivative.

Worked solution

$f''(x) = 2 > 0$ for all x , hence f is strictly convex. Therefore any critical point is the unique global minimum.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- A local extremum need not be global.
- Convex and increasing are unrelated properties.
- For nonsmooth functions, convexity can be studied without derivatives.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Is e^x increasing and convex?
- 2 Find the global minimum of $(x - 2)^2 + 1$.
- 3 Describe the epigraph of the constant function $f(x) = 3$.

Concise solutions

- ① Yes: $f' = f'' = e^x > 0$.
- ② Minimum value 1 at $x = 2$.
- ③ The half-plane $\{(x, t) : t \geq 3\}$.

Chapter 4

Trigonometry

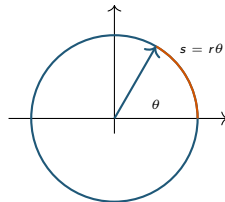
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Angles, degrees and radians — overview

Concept

Angles may be measured in degrees or radians. A radian is defined through arc length on a circle.

- $180^\circ = \pi$ radians.
- $360^\circ = 2\pi$ radians.
- On a circle of radius r , arc length satisfies $s = r\theta$ when θ is in radians.



Angles, degrees and radians — example

Problem

Convert 150° to radians and $5\pi/6$ radians to degrees.

Worked solution

$150^\circ = 150\pi/180 = 5\pi/6$. Conversely $(5\pi/6)(180/\pi) = 150^\circ$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not use degree-mode identities inside calculus formulas.
- Cancel units explicitly when converting.
- Angles differing by $2k\pi$ have the same direction.

Angles, degrees and radians — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Convert 45° to radians.
- 2 Convert $7\pi/4$ to degrees.
- 3 How many radians are in a quarter turn?

Concise solutions

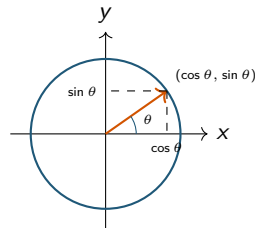
- ① $\pi/4$.
- ② 315° .
- ③ $\pi/2$.

The unit circle and sine and cosine — overview

Concept

On the unit circle, the point reached by angle θ has coordinates $(\cos \theta, \sin \theta)$. This geometric definition explains signs, periodicity and the identity $\sin^2 \theta + \cos^2 \theta = 1$.

- $-1 \leq \sin \theta, \cos \theta \leq 1$.
- Both sine and cosine have period 2π .
- $\cos$ is even and $\sin$ is odd.



The unit circle and sine and cosine — example

Problem

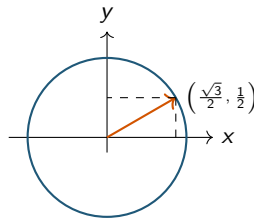
Read off $\cos \theta$ and $\sin \theta$ when $\theta = \pi/6$.

Worked solution

At $\theta = \pi/6$, the point on the unit circle is

$$\left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right).$$

Hence $\cos(\pi/6) = \sqrt{3}/2$ and $\sin(\pi/6) = 1/2$.



The unit circle and sine and cosine — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Memorize standard-angle values through the unit circle, not as isolated numbers.
- Check the quadrant before assigning signs.
- Coordinates are $(\cos, \sin)$, not the reverse.

The unit circle and sine and cosine — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Evaluate $\sin(3\pi/2)$.
- 2 Evaluate $\cos(7\pi/4)$.
- 3 Verify $\sin^2(\pi/3) + \cos^2(\pi/3) = 1$.

The unit circle and sine and cosine — solutions

Concise solutions

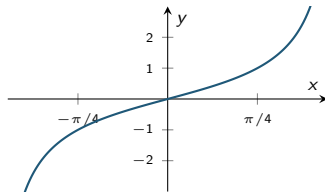
- ① -1 .
- ② $\sqrt{2}/2$.
- ③ $3/4 + 1/4 = 1$.

Tangent and the main trigonometric identities — overview

Concept

The tangent is defined by $\tan x = \sin x / \cos x$ whenever $\cos x \neq 0$. The main identities connect sine, cosine and tangent.

- $\sin^2 x + \cos^2 x = 1$.
- $1 + \tan^2 x = 1 / \cos^2 x$.
- Tangent has vertical asymptotes where $\cos x = 0$.



Tangent and the main trigonometric identities — example

Problem

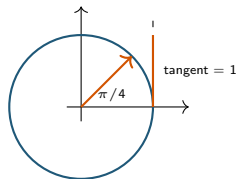
Compute $\tan(\pi/4)$ and verify $1 + \tan^2(\pi/4) = 1/\cos^2(\pi/4)$.

Worked solution

$\sin(\pi/4) = \cos(\pi/4) = \sqrt{2}/2$, so $\tan(\pi/4) = 1$. Then

$$1 + \tan^2(\pi/4) = 2, \quad \frac{1}{\cos^2(\pi/4)} = \frac{1}{1/2} = 2.$$

The identity is verified.



Tangent and the main trigonometric identities — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Square-root recovery always requires a sign decision.
- Tangent is not defined when cosine vanishes.
- Do not replace $\tan(a + b)$ by $\tan a + \tan b$.

Tangent and the main trigonometric identities — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $\cos \theta = 12/13$ in quadrant IV, find $\sin \theta$.
- 2 Evaluate $\tan(\pi/4)$.
- 3 Solve $\tan \theta = 0$.

Tangent and the main trigonometric identities — solutions

Concise solutions

- ① $-5/13$.
- ② 1 .
- ③ $\theta = k\pi, k \in \mathbb{Z}$.

Addition, subtraction and double-angle formulas — overview

Concept

Angle-addition formulas allow exact values at nonstandard angles and are the source of double-angle identities.

- $\sin(a + b) = \sin a \cos b + \cos a \sin b.$
- $\cos(a + b) = \cos a \cos b - \sin a \sin b.$
- $\sin 2x = 2 \sin x \cos x.$
- $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1.$

Addition, subtraction and double-angle formulas — example

Problem

Compute $\cos(75^\circ)$ exactly.

Worked solution

$$75^\circ = 45^\circ + 30^\circ, \text{ so } \cos 75^\circ = \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \frac{1}{2} = \frac{\sqrt{6}-\sqrt{2}}{4}.$$

Addition, subtraction and double-angle formulas — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- The sign in the cosine addition formula is minus.
- Choose the double-angle version that best matches the available information.
- Keep exact radicals until the end.

Addition, subtraction and double-angle formulas — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $\sin 75^\circ$.
- 2 Express $\cos 2x$ using only $\sin x$.
- 3 If $\sin x = \cos x > 0$, find $\sin 2x$.

Addition, subtraction and double-angle formulas — solutions

Concise solutions

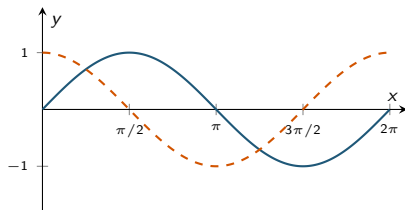
- ① $(\sqrt{6} + \sqrt{2})/4$.
- ② $1 - 2 \sin^2 x$.
- ③ 1.

Trigonometric graphs and transformations — overview

Concept

Sine and cosine are periodic waves. Amplitude controls vertical stretch, period controls horizontal repetition, and added constants shift the graph.

- $A \sin(Bx + C) + D$ has amplitude $|A|$ and midline $y = D$.
- The period is $2\pi/|B|$.
- The phase shift comes from rewriting $Bx + C = B(x - x_0)$.



Trigonometric graphs and transformations — example

Problem

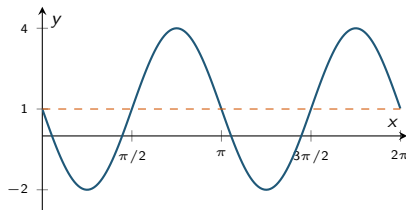
Describe $f(x) = 3 \sin(2x - \pi) + 1$.

Worked solution

Amplitude 3, period π , vertical shift +1, and

$$2x - \pi = 2\left(x - \frac{\pi}{2}\right)$$

so the graph is shifted right by $\pi/2$. Its midline is $y = 1$.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not call ϕ itself the phase shift unless $\omega = 1$.
- Amplitude is nonnegative.
- Period and horizontal scaling are reciprocally related.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the period of $\cos(5x)$.
- 2 Find the amplitude and vertical shift of $-2 \sin x + 4$.
- 3 Write $\sin(3x + 6)$ with the horizontal shift visible.

Concise solutions

- ① $2\pi/5$.
- ② Amplitude 2, shift +4.
- ③ $\sin(3(x + 2))$, so shift left by 2.

Concept

Since sine, cosine and tangent are not one-to-one on their full natural domains, inverse functions are defined after restricting them to principal intervals.

- $\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2]$.
- $\arccos : [-1, 1] \rightarrow [0, \pi]$.
- $\arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$.

Inverse trigonometric functions — example

Problem

Evaluate $\arcsin(-1/2)$ and $\arccos(-1/2)$.

Worked solution

The principal values are $\arcsin(-1/2) = -\pi/6$ and $\arccos(-1/2) = 2\pi/3$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Inverse trigonometric functions return principal values.
- $\sin(\arcsin x) = x$, but $\arcsin(\sin x)$ need not equal x outside the principal interval.
- Do not confuse $\sin^{-1} x$ with $1/\sin x$.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Evaluate $\arctan 1$.
- 2 Evaluate $\arccos 0$.
- 3 Compute $\sin(\arccos(3/5))$.

Concise solutions

- ① $\pi/4$.
- ② $\pi/2$.
- ③ $4/5$, taking the nonnegative sine implied by the range of $\arccos$.

Concept

Trigonometric equations typically have infinitely many solutions because of periodicity. The general solution must include all periods, not only principal solutions.

- $\sin x = 0 \Rightarrow x = k\pi.$
- $\cos x = 0 \Rightarrow x = \pi/2 + k\pi.$
- $\tan x = a \Rightarrow x = \arctan a + k\pi.$

Trigonometric equations and inequalities — example

Problem

Solve $\sin x = \sqrt{3}/2$ on $\mathbb{R}$.

Worked solution

Within $[0, 2\pi)$ the solutions are $\pi/3$ and $2\pi/3$. Therefore $x = \pi/3 + 2k\pi$ or $x = 2\pi/3 + 2k\pi$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not stop after the inverse-function value.
- Use the unit circle to find all solutions in one period.
- For inequalities, track intervals where the graph lies above or below the threshold.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Solve $\cos x = 1$.
- 2 Solve $\tan x = -1$.
- 3 Solve $\sin x > 0$ on $[0, 2\pi)$.

Concise solutions

- ① $x = 2k\pi$.
- ② $x = -\pi/4 + k\pi$.
- ③ $(0, \pi)$.

Concept

In radians, differentiation and integration preserve the sine–cosine cycle. These formulas later support Fourier methods, oscillatory models and many substitutions.

- $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$.
- $(\tan x)' = \sec^2 x$ where defined.
- $\int \cos x \, dx = \sin x + C$, $\int \sin x \, dx = -\cos x + C$.

Trigonometric derivatives and antiderivatives — example

Problem

Differentiate $f(x) = \sin(3x^2)$.

Worked solution

By the chain rule, $f'(x) = \cos(3x^2) \cdot 6x = 6x \cos(3x^2)$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- The chain rule contributes the derivative of the inner argument.
- Signs in sine/cosine antiderivatives are easy to reverse.
- Derivative formulas presume radian measure.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Differentiate $\cos(5x)$.
- 2 Integrate $\cos(4x)$.
- 3 Differentiate $\tan(x^2)$.

Concise solutions

- ① $-5 \sin(5x)$.
- ② $\frac{1}{4} \sin(4x) + C$.
- ③ $2x \sec^2(x^2)$.

Chapter 5

Sequences, limits and continuity

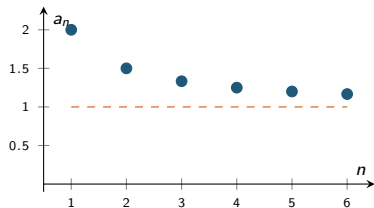
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Sequences and their graphs — overview

Concept

A real sequence is a function $a : \mathbb{N} \rightarrow \mathbb{R}$, usually written (a_n) . The index n is discrete even when the formula resembles a function of a real variable.

- A sequence may be defined explicitly, recursively, or algorithmically.
- Monotonicity and boundedness are useful qualitative properties.
- Subsequences retain selected indices n_k with $n_1 < n_2 < \dots$.



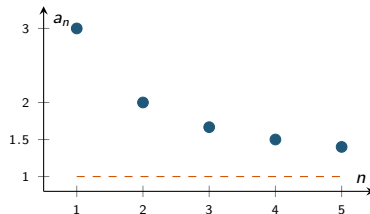
Sequences and their graphs — example

Problem

For $a_n = 1 + \frac{2}{n}$, compute the first four terms and describe the trend.

Worked solution

$a_1 = 3$, $a_2 = 2$, $a_3 = 5/3$, $a_4 = 3/2$. The sequence decreases toward 1.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

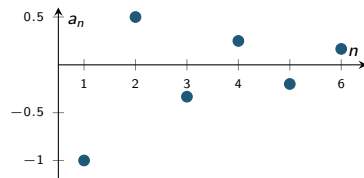
- Do not treat the index as a continuous variable unless doing so is an auxiliary technique.
- A recursive rule needs an initial value.
- Convergence concerns tail behavior, not the first few terms.

Sequences and their graphs — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute the first five terms of $a_n = (-1)^n/n$.
- 2 Is $a_n = n^2$ bounded?
- 3 Give a subsequence of $a_n = (-1)^n$ that is constantly 1.



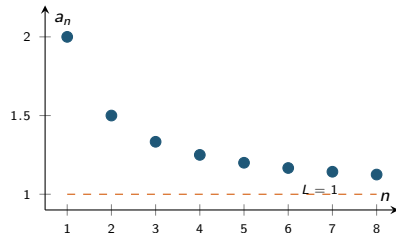
Concise solutions

- ① $-1, 1/2, -1/3, 1/4, -1/5$ if n starts at 1.
- ② No, it is unbounded above.
- ③ $a_{2k} = 1$.

Concept

The statement $a_n \rightarrow L$ means that sequence terms can be made arbitrarily close to L by taking n sufficiently large.

- Formally: for every $\varepsilon > 0$ there exists N such that $n \geq N \Rightarrow |a_n - L| < \varepsilon$.
- A convergent sequence has a unique limit.
- Every convergent real sequence is bounded.



Limits of sequences — example

Problem

Show directly that $1/n \rightarrow 0$.

Worked solution

Given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then $n \geq N$ implies $|1/n| = 1/n \leq 1/N < \varepsilon$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- N may depend on ε , but not on n after it is chosen.
- Small early terms do not matter; the condition is eventually true.
- Boundedness alone does not imply convergence.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find $\lim_{n \rightarrow \infty} (3 + 2/n)$.
- 2 Does $(-1)^n$ converge?
- 3 Find $\lim n/(n+1)$.

Concise solutions

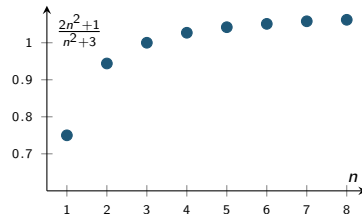
- ① 3.
- ② No; even and odd subsequences have different limits.
- ③ 1.

Algebra of limits and dominant terms — overview

Concept

Finite limits can be combined through sums, products and quotients, provided denominators do not converge to zero. For rational expressions in n , dominant powers often determine the limit.

- If $a_n \rightarrow a$, $b_n \rightarrow b$, then $a_n + b_n \rightarrow a + b$.
- $a_n b_n \rightarrow ab$.
- $a_n / b_n \rightarrow a/b$ when $b \neq 0$.



Algebra of limits and dominant terms — example

Problem

Compute $\lim_{n \rightarrow \infty} \frac{3n^2 - n + 1}{2n^2 + 5n}$.

Worked solution

Divide numerator and denominator by n^2 : the lower-order terms vanish, giving $3/2$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Never substitute ∞ as if it were a real number.
- Quotient rules require a nonzero limiting denominator.
- Dominant-term reasoning should be made explicit by normalization.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find $\lim(5n + 1)/(n - 7)$.
- 2 Find $\lim n/(n^2 + 1)$.
- 3 Find $\lim(2 + 1/n)^2$.

Concise solutions

- ① 5.
- ② 0.
- ③ 4.

Concept

The limit $\lim_{x \rightarrow x_0} f(x) = L$ describes values of $f(x)$ near x_0 , not necessarily the value at x_0 . One-sided limits distinguish approach from the left and right.

- A two-sided finite limit exists iff the two one-sided limits exist and agree.
- The point x_0 need not belong to the domain.
- Limits are local statements.

Function limits and one-sided limits — example

Problem

Let $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$. Find $\lim_{x \rightarrow 1} f(x)$.

Worked solution

For $x \neq 1$, $f(x) = x + 1$. Hence the limit is 2, even though the original formula is undefined at 1.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not replace a limit by direct substitution when substitution gives an indeterminate form.
- A missing function value does not prevent a limit from existing.
- Left and right behavior must agree for a two-sided limit.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find $\lim_{x \rightarrow 2} (x^2 + 1)$.
- 2 Find $\lim_{x \rightarrow 0^+} 1/x$.
- 3 Does $\lim_{x \rightarrow 0} |x|/x$ exist?

Concise solutions

- ① 5.
- ② $+\infty$.
- ③ No: left limit is -1 , right limit is 1 .

Indeterminate forms and algebraic techniques — overview

Concept

Expressions such as $0/0$, ∞/∞ , $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 and ∞^0 are indeterminate forms: they signal that further analysis is required.

- Factorization resolves many polynomial $0/0$ limits.
- Rationalization is useful with square roots.
- Common denominators resolve some $\infty - \infty$ forms.

Indeterminate forms and algebraic techniques — example

Problem

Compute $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{x}$.

Worked solution

Multiply by the conjugate: $\frac{1}{\sqrt{1+x}+1}$, so the limit is $1/2$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- An indeterminate form is not the answer.
- Algebraic transformations must preserve the expression near the limiting point.
- Do not use l'Hôpital before checking its hypotheses.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $\lim_{x \rightarrow 1} (x^2 - 1)/(x - 1)$.
- 2 Compute $\lim_{x \rightarrow \infty} (2x^2 + 1)/(x^2 - 3)$.
- 3 Rationalize $(\sqrt{x + 4} - 2)/x$ near 0.

Concise solutions

- ① 2.
- ② 2.
- ③ It becomes $1/(\sqrt{x+4} + 2)$, so the limit is $1/4$.

Concept

Several standard limits are building blocks for derivatives, Taylor expansions and continuous compounding.

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$
- $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1.$
- $\lim_{n \rightarrow \infty} (1 + a/n)^n = e^a.$

Notable limits and exponential asymptotics — example

Problem

Compute $\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$.

Worked solution

Write $\frac{\sin(5x)}{x} = 5 \frac{\sin(5x)}{5x}$. The second factor tends to 1, so the limit is 5.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Arguments must be normalized to the notable-limit form.
- The sine limit is valid in radians.
- e^a arises from the entire sequence $(1 + a/n)^n$, not from termwise substitution.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find $\lim_{x \rightarrow 0} (e^{3x} - 1)/x$.
- 2 Find $\lim_{x \rightarrow 0} \log(1 + 2x)/x$.
- 3 Find $\lim_{n \rightarrow \infty} (1 + 0.04/n)^n$.

Concise solutions

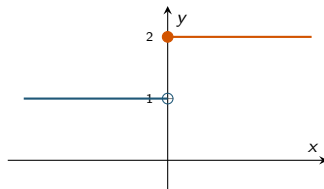
- ① 3.
- ② 2.
- ③ $e^{0.04}$.

Continuity and discontinuities — overview

Concept

A function is continuous at x_0 when $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. Discontinuities are classified by the behavior of one-sided limits.

- Removable: a finite limit exists but the value is missing or different.
- Jump: finite one-sided limits exist but differ.
- Second-kind: at least one one-sided limit fails to exist finitely.



Continuity and discontinuities — example

Problem

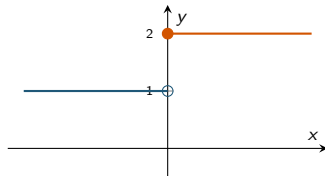
Classify the discontinuity of

$$f(x) = \begin{cases} 1, & x < 0, \\ 2, & x \geq 0. \end{cases}$$

at $x = 0$.

Worked solution

The left-hand limit is 1 and the right-hand limit is 2, so the one-sided limits are finite but different. Therefore $x = 0$ is a jump discontinuity.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Continuity is checked pointwise.
- A graph without a visible break can still require domain checks.
- Piecewise functions require one-sided analysis at junctions.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Is $1/x$ continuous at 0?
- 2 Where is $\sqrt{x}$ continuous?
- 3 Find c making $f(x) = x + 1$ for $x < 0$ and $f(x) = c$ for $x \geq 0$ continuous at 0.

Concise solutions

- ① No; it is not defined there.
- ② On $[0, \infty)$ in its natural domain.
- ③ $c = 1$.

Concept

Continuity on closed bounded intervals gives strong existence results. Asymptotic notation compares small and large quantities near a point or at infinity.

- Weierstrass: a continuous function on $[a, b]$ attains a maximum and a minimum.
- Intermediate value theorem: continuous functions take every value between endpoint values.
- $f = o(g)$ means $f/g \rightarrow 0$; $f \sim g$ means $f/g \rightarrow 1$.

Theorems on continuous functions, infinitesimals and infinities — example

Problem

Use the intermediate value theorem to show $x^3 + x - 1 = 0$ has a root in $(0, 1)$.

Worked solution

$f(0) = -1$ and $f(1) = 1$. Since f is continuous, it takes the value 0 at some $c \in (0, 1)$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Existence theorems do not automatically give a formula for the solution.
- Closed and bounded interval hypotheses matter for guaranteed extrema.
- $f \sim g$ is stronger than merely having the same sign or order of magnitude.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Why does e^x attain no maximum on $\mathbb{R}$ despite being continuous?
- 2 Show $x^2 = o(x)$ as $x \rightarrow 0$.
- 3 Give an asymptotic equivalent of $\sin x$ as $x \rightarrow 0$.

Concise solutions

- ① The domain is not a compact interval and the function is unbounded.
- ② $x^2/x = x \rightarrow 0$.
- ③ $\sin x \sim x$.

Chapter 6

Differential calculus and optimization

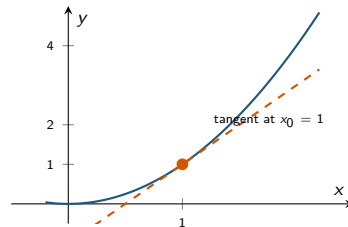
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Derivative, one-sided derivatives and geometric meaning — overview

Concept

The derivative of f at x_0 is the limit of the difference quotient. It measures instantaneous rate of change and the slope of the tangent line.

- $f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$.
- Left and right derivatives must agree for differentiability.
- Differentiability implies continuity, but continuity does not imply differentiability.



Derivative, one-sided derivatives and geometric meaning — example

Problem

Compute the derivative of $f(x) = x^2$ at x_0 from the definition.

Worked solution

$$\frac{(x_0+h)^2 - x_0^2}{h} = 2x_0 + h \rightarrow 2x_0. \text{ Thus } f'(x_0) = 2x_0.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not cancel h before expanding when the numerator is not factorized.
- A corner can be continuous but nondifferentiable.
- The tangent slope is local; it does not describe the entire graph.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $(x^3)'$ from known rules.
- 2 Is $|x|$ differentiable at 0?
- 3 Find the tangent line to x^2 at $x = 1$.

Concise solutions

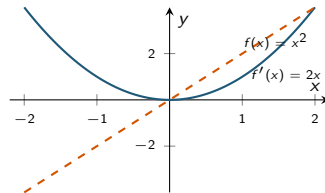
- ① $3x^2$.
- ② No: left derivative -1 , right derivative 1 .
- ③ $y - 1 = 2(x - 1)$, so $y = 2x - 1$.

Derivative rules and elementary derivatives — overview

Concept

Derivative rules convert complicated expressions into combinations of elementary derivatives.

- $(f + g)' = f' + g'$.
- $(fg)' = f'g + fg'$.
- $(f/g)' = (f'g - fg')/g^2$ where $g \neq 0$.
- $(e^x)' = e^x$, $(\log x)' = 1/x$ for $x > 0$.



Derivative rules and elementary derivatives — example

Problem

Differentiate $f(x) = x^2 e^x$.

Worked solution

Product rule gives $f'(x) = 2xe^x + x^2 e^x = e^x(x^2 + 2x)$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not differentiate products term by term as $f'g'$.
- Keep domain restrictions for quotients and logarithms.
- Factor common terms after differentiation to simplify sign analysis.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Differentiate $(x^2 + 1)/(x - 1)$.
- 2 Differentiate $x \log x$.
- 3 Differentiate e^{2x} .

Concise solutions

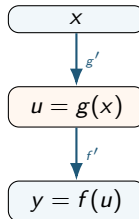
- ① $[(2x)(x-1) - (x^2 + 1)]/(x-1)^2$.
- ② $\log x + 1$ for $x > 0$.
- ③ $2e^{2x}$.

Chain rule and derivatives of composite functions — overview

Concept

If $y = f(g(x))$, then the derivative is the derivative of the outer function evaluated at the inner function, multiplied by the derivative of the inner function.

- $(f \circ g)'(x) = f'(g(x))g'(x)$.
- Nested compositions require repeated application.
- Domain restrictions of the composition remain essential.



multiply the rates: $(f \circ g)' = f'(g(x))g'(x)$

Chain rule and derivatives of composite functions — example

Problem

Differentiate $\log(1 + x^2)$.

Worked solution

Outer derivative: $1/u$; inner derivative: $2x$. Therefore $f'(x) = 2x/(1 + x^2)$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not omit the derivative of the inner function.
- Identify layers before differentiating.
- Simplify only after the chain rule is correctly applied.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Differentiate $\sqrt{1 + 3x}$.
- 2 Differentiate $e^{x^2 - x}$.
- 3 Differentiate $\sin(\log x)$.

Concise solutions

- ① $3/[2\sqrt{1+3x}]$.
- ② $(2x-1)e^{x^2-x}$.
- ③ $\cos(\log x)/x$, for $x > 0$.

Concept

Logarithmic differentiation is useful for products, powers with variable exponents and elasticity calculations. The elasticity of f with respect to x is $E_f(x) = xf'(x)/f(x)$ when defined.

- Differentiate $\log f(x)$ to convert products into sums.
- For $f(x) = Cx^\alpha$, elasticity is constant and equal to α .
- Elasticity is dimensionless.

Logarithmic differentiation and elasticities — example

Problem

Differentiate $y = x^x$ for $x > 0$.

Worked solution

Take logs: $\log y = x \log x$. Then $y'/y = \log x + 1$, so $y' = x^x(\log x + 1)$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Logarithmic differentiation requires positive quantities unless absolute values/complex branches are handled.
- Elasticity is not the same as the derivative.
- Do not forget to multiply back by y after differentiating $\log y$.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the elasticity of $f(x) = 5x^{3/2}$.
- 2 Differentiate $y = (x^2 + 1)^x$ for $x > 0$.
- 3 Interpret elasticity 2 locally.

Concise solutions

- ① $3/2$.
- ② $y' = (x^2 + 1)^x [\log(x^2 + 1) + 2x^2/(x^2 + 1)]$.
- ③ A 1% increase in x produces approximately a 2% increase in f locally.

Concept

Higher derivatives measure changes in lower-order rates. The second derivative is especially important for curvature, convexity and second-order approximations.

- $f'' = (f')'$.
- If $f'' > 0$ on an interval, f is strictly convex there.
- If $f'' < 0$, f is strictly concave there.

Higher derivatives and local curvature — example

Problem

For $f(x) = x^4 - 2x^2$, compute $f''(x)$ and identify where the graph is locally convex.

Worked solution

$f'(x) = 4x^3 - 4x$ and $f''(x) = 12x^2 - 4$. Convexity requires $12x^2 - 4 > 0$, i.e. $|x| > 1/\sqrt{3}$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- A zero second derivative alone does not prove an inflection point.
- Curvature can change only after checking sign on both sides.
- Higher derivatives require sufficient differentiability.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute the third derivative of e^{2x} .
- 2 Find where x^3 is concave.
- 3 Does x^4 have an inflection point at 0?

Concise solutions

- ① $8e^{2x}$.
- ② $f''(x) = 6x < 0$ for $x < 0$.
- ③ No; $f'' = 12x^2 \geq 0$ does not change sign.

Concept

The mean value theorem connects an average rate of change on an interval to an instantaneous derivative at an interior point.

- Rolle: if $f(a) = f(b)$ and f is continuous/differentiable as required, some c satisfies $f'(c) = 0$.
- Lagrange MVT: $f'(c) = [f(b) - f(a)]/(b - a)$.
- If $f' = 0$ on an interval, then f is constant there.

Rolle, mean value theorem and consequences — example

Problem

Apply the mean value theorem to $f(x) = x^2$ on $[1, 3]$.

Worked solution

The secant slope is $(9 - 1)/(3 - 1) = 4$. Since $f'(x) = 2x$, set $2c = 4$, giving $c = 2$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Check continuity on the closed interval and differentiability inside.
- The theorem guarantees existence, not uniqueness.
- Do not use endpoint derivatives as the MVT point.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the Rolle point for $f(x) = x^2 - 4x$ on $[0, 4]$.
- 2 What does $f'(x) > 0$ imply on an interval?
- 3 Can MVT be applied to $|x|$ on $[-1, 1]$?

Concise solutions

- ① $c = 2$.
- ② f is strictly increasing.
- ③ No, because $|x|$ is not differentiable at 0.

l'Hôpital's rule — overview

Concept

Under suitable differentiability and limit conditions, l'Hôpital's rule evaluates certain $0/0$ or ∞/∞ limits by differentiating numerator and denominator.

- Use only after identifying an eligible indeterminate quotient.
- The transformed quotient is f'/g' , not the derivative of f/g .
- Repeated application is possible when conditions remain valid.

l'Hôpital's rule — example

Problem

Compute $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

Worked solution

First application gives $\lim(e^x - 1)/(2x)$, still $0/0$. A second gives $\lim e^x/2 = 1/2$.

l'Hôpital's rule — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not apply l'Hôpital to a non-indeterminate finite quotient.
- Products and differences must first be rewritten into suitable forms.
- Algebra or Taylor expansion may be simpler.

l'Hôpital's rule — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $\lim_{x \rightarrow 0} \sin x / x$ using l'Hôpital.
- 2 Compute $\lim_{x \rightarrow \infty} \log x / x$.
- 3 Why is l'Hôpital not needed for $\lim_{x \rightarrow 1} (x + 1) / (x + 2)$?

l'Hôpital's rule — solutions

Concise solutions

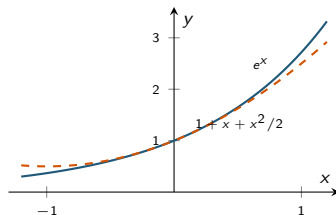
- ① 1.
- ② 0.
- ③ Direct substitution gives the determinate value $2/3$.

Taylor polynomials and local approximation — overview

Concept

Taylor polynomials approximate a smooth function near a base point by matching derivatives up to a chosen order.

- The linear Taylor approximation is the tangent line.
- Quadratic terms capture local curvature.
- The approximation is best near the expansion point.



Taylor polynomials and local approximation — example

Problem

Find the second-order Taylor polynomial of e^x at 0.

Worked solution

All derivatives of e^x equal e^x , hence at 0 they equal 1: $T_2(x) = 1 + x + x^2/2$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Taylor approximation is local unless an error bound is controlled.
- Factorials belong in every coefficient.
- Choose the expansion point near the value of interest.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find T_2 for $\log(1+x)$ at 0.
- 2 Approximate $e^{0.1}$ with T_2 .
- 3 Find the first-order approximation to $\sqrt{x}$ at $x = 1$.

Concise solutions

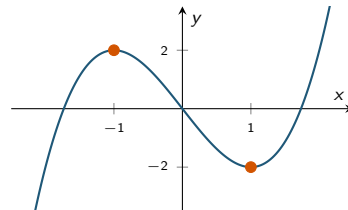
- ① $x - x^2/2.$
- ② $1 + 0.1 + 0.005 = 1.105.$
- ③ $1 + \frac{1}{2}(x - 1).$

Monotonicity and first-derivative optimization — overview

Concept

Critical points are candidates for local extrema. A sign chart for f' determines increasing/decreasing intervals and classifies many extrema.

- Critical points occur where $f' = 0$ or f' does not exist inside the domain.
- f' changing $+$ $\rightarrow$ $-$ gives a local maximum.
- f' changing $-$ $\rightarrow$ $+$ gives a local minimum.



Monotonicity and first-derivative optimization — example

Problem

Study the extrema of $f(x) = x^3 - 3x$.

Worked solution

$f'(x) = 3(x^2 - 1)$, so critical points are ± 1 . The sign pattern is $+, -, +$, hence a local maximum at $x = -1$ and local minimum at $x = 1$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Solving $f' = 0$ only finds candidates.
- Endpoints matter on constrained closed intervals.
- Nondifferentiable points may be extrema.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the extrema of $x^2 - 4x + 7$.
- 2 Find the maximum of $x(10 - x)$ on $[0, 10]$.
- 3 Can $|x|$ have a minimum despite no derivative at 0?

Concise solutions

- ① Unique minimum at $x = 2$, value 3.
- ② At $x = 5$, value 25.
- ③ Yes, global minimum 0 at $x = 0$.

Concept

Second-derivative information organizes curvature and strengthens optimization conclusions. On a convex domain, a differentiable convex function has no spurious local minima.

- $f'' > 0$ implies strict convexity.
- A sign change of f'' indicates an inflection point when the function is suitably regular.
- For strictly convex f , a critical point is the unique global minimum.

Convexity, concavity, inflection points and global optimization — example

Problem

Find the global minimum of $f(x) = e^x - x$.

Worked solution

$f'(x) = e^x - 1$, so $x = 0$ is the only critical point. Since $f''(x) = e^x > 0$, f is strictly convex and $x = 0$ is the unique global minimum, with value 1.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- $f''(x_0) = 0$ is only a candidate for an inflection point.
- Convexity can be global even when slopes are negative in part of the domain.
- Constraint sets change what counts as a feasible optimum.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Show $-\log x$ is convex on $(0, \infty)$.
- 2 Find inflection points of $x^3 - 3x^2$.
- 3 Why is a local minimum of a convex function global?

Concise solutions

- ① $f''(x) = 1/x^2 > 0$.
- ② $f'' = 6x - 6$, sign changes at $x = 1$.
- ③ Convexity prevents the graph from dipping below a local supporting level elsewhere.

Chapter 7

Integrals and integration techniques

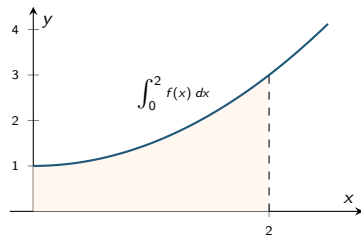
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Definite integrals and signed area — overview

Concept

The definite integral $\int_a^b f(x) dx$ measures signed area between the graph of f and the x -axis on $[a, b]$.

- Area above the axis contributes positively.
- Area below the axis contributes negatively.
- The integral combines geometry and accumulation.



Definite integrals and signed area — example

Problem

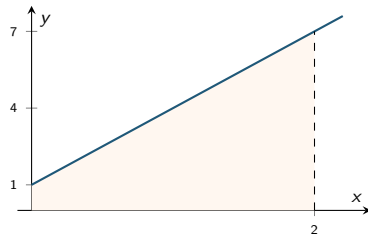
Compute $\int_0^2 (3x + 1) dx$.

Worked solution

An antiderivative is $\frac{3}{2}x^2 + x$. Therefore

$$\int_0^2 (3x + 1) dx = \left[\frac{3}{2}x^2 + x \right]_0^2 = 6 + 2 = 8.$$

The graph also shows the positive area under the line on $[0, 2]$.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Signed area differs from total geometric area when f changes sign.
- Keep bounds attached to the variable of integration.
- An integral is not obtained by substituting endpoints into the integrand.

Definite integrals and signed area — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $\int_0^1 x \, dx$.
- 2 Use symmetry to evaluate $\int_{-2}^2 x^3 \, dx$.
- 3 Rewrite $\int_3^1 f(x) \, dx$ with increasing bounds.

Concise solutions

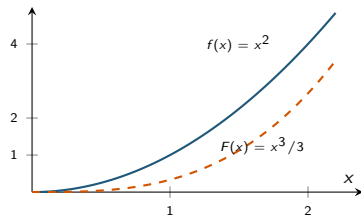
- ① $1/2$.
- ② 0 because the integrand is odd on a symmetric interval.
- ③ $-\int_1^3 f(x) dx$.

Primitive functions and the fundamental theorem of calculus — overview

Concept

A primitive F of f satisfies $F' = f$. The fundamental theorem connects accumulation and differentiation and turns many definite integrals into endpoint evaluations.

- $\int f(x) dx = F(x) + C$.
- If $F' = f$, then $\int_a^b f(x) dx = F(b) - F(a)$.
- Any two primitives on an interval differ by a constant.



Primitive functions and the fundamental theorem of calculus — example

Problem

Find all primitives of $f(x) = 4x^3 - 2x$.

Worked solution

Integrate term by term: $F(x) = x^4 - x^2 + C$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- The constant C is mandatory for indefinite integrals.
- Do not add C to a definite integral after endpoint evaluation.
- A primitive formula may be valid only on intervals avoiding singularities.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Integrate x^5 .
- 2 Compute $\int_1^2 1/x \, dx$.
- 3 Differentiate $F(x) = \int_0^x e^{-t^2} \, dt$.

Concise solutions

- ① $x^6/6 + C$.
- ② $\log 2$.
- ③ $F'(x) = e^{-x^2}$ by the fundamental theorem.

Concept

Integration is linear, so known elementary primitives can be combined. A small reliable table is more useful than memorizing many unrelated special cases.

- $\int x^\alpha dx = x^{\alpha+1}/(\alpha+1) + C$ for $\alpha \neq -1$.
- $\int dx/x = \log|x| + C$.
- $\int e^{ax} dx = e^{ax}/a + C$ for $a \neq 0$.
- $\int \cos(ax) dx = \sin(ax)/a + C$.

Basic antiderivative table and linearity — example

Problem

Compute $\int (3x^2 - 4/x + 2e^{2x}) dx$.

Worked solution

$$x^3 - 4 \log |x| + e^{2x} + C.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- The power rule fails at exponent -1 .
- Use $\log |x|$ for $1/x$ on intervals not crossing zero.
- Linear coefficients from inner derivatives affect primitives.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Integrate x^{-2} .
- 2 Integrate $5 \sin(3x)$.
- 3 Integrate $2/(1 + x^2)$.

Basic antiderivative table and linearity — solutions

Concise solutions

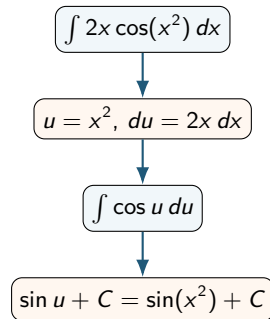
- ① $-1/x + C.$
- ② $-(5/3) \cos(3x) + C.$
- ③ $2 \arctan x + C.$

Substitution: the reverse chain rule — overview

Concept

Substitution recognizes a composite derivative. If $u = g(x)$, then $du = g'(x) dx$ and the integral is rewritten entirely in terms of u .

- Look for an inner expression together with its derivative.
- After substitution, no stray x should remain.
- For definite integrals, either transform the bounds or substitute back before evaluating.



Substitution: the reverse chain rule — example

Problem

Compute $\int 2xe^{x^2} dx$.

Worked solution

Set $u = x^2$, so $du = 2x dx$. Then $\int e^u du = e^u + C = e^{x^2} + C$.

Substitution: the reverse chain rule — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- A substitution is not merely renaming part of an expression.
- Scale constants correctly when the derivative differs by a factor.
- Do not mix old x -bounds with a new u -integrand.

Substitution: the reverse chain rule — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Integrate $\int x/(1+x^2) dx$.
- 2 Integrate $\int \cos(5x) dx$ by substitution.
- 3 Compute $\int_0^1 2x(1+x^2)^3 dx$.

Substitution: the reverse chain rule — solutions

Concise solutions

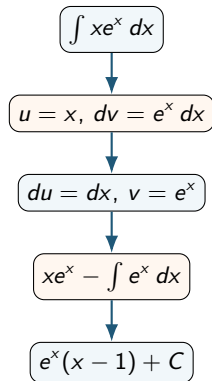
- ① $\frac{1}{2} \log(1 + x^2) + C.$
- ② $\frac{1}{5} \sin(5x) + C.$
- ③ With $u = 1 + x^2$, bounds 1 to 2: $\int_1^2 u^3 du = 15/4.$

Integration by parts — overview

Concept

Integration by parts reverses the product rule: $\int u dv = uv - \int v du$. It is especially useful for products of algebraic, logarithmic, exponential and trigonometric factors.

- Choose u to become simpler when differentiated.
- Choose dv so it can be integrated reliably.
- For definite integrals, evaluate the boundary term uv at both endpoints.



Integration by parts — example

Problem

Compute $\int x e^x dx$.

Worked solution

Take $u = x$, $dv = e^x dx$. Then $du = dx$, $v = e^x$, hence $\int x e^x dx = x e^x - \int e^x dx = e^x(x - 1) + C$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not forget the minus sign.
- Both u and dv must be chosen from the original integrand.
- Repeated parts is often required for polynomial times exponential/trigonometric functions.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Integrate $\int x \cos x \, dx$.
- 2 Integrate $\int \log x \, dx$.
- 3 Compute $\int_0^1 x e^x \, dx$.

Concise solutions

① $x \sin x + \cos x + C.$

② $x \log x - x + C.$

③ $[e^x(x-1)]_0^1 = 1.$

Repeated integration by parts — overview

Concept

When a polynomial multiplies an exponential or sine/cosine, integration by parts can be repeated until the polynomial disappears.

- Tabular integration is a bookkeeping shortcut for repeated parts.
- Differentiate the polynomial; integrate the exponential/trigonometric factor.
- Alternating signs arise from the formula.

Repeated integration by parts — example

Problem

Compute $\int x^2 e^x dx$.

Worked solution

First parts gives $x^2 e^x - 2 \int x e^x dx$. Using the previous result, this is $e^x(x^2 - 2x + 2) + C$.

Repeated integration by parts — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Stopping after one application may leave an integral of the same type.
- Check the final expression by differentiation.
- Keep constants factored to reduce algebraic errors.

Repeated integration by parts — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Integrate $\int x^2 \cos x \, dx$.
- 2 Integrate $\int x^3 e^x \, dx$.
- 3 Differentiate $e^x(x^2 - 2x + 2)$ to verify the example.

Repeated integration by parts — solutions

Concise solutions

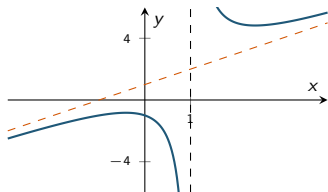
- ① $x^2 \sin x + 2x \cos x - 2 \sin x + C.$
- ② $e^x(x^3 - 3x^2 + 6x - 6) + C.$
- ③ The derivative simplifies to $x^2 e^x.$

Rational functions and polynomial division — overview

Concept

A rational function is a quotient of polynomials. When the numerator degree is at least the denominator degree, polynomial division separates a polynomial part from a proper rational remainder.

- Example: $(x^2 + 1)/(x - 1) = x + 1 + 2/(x - 1)$.
- The remainder term reveals vertical asymptotes.
- The polynomial quotient reveals oblique or horizontal behavior.



Rational functions and polynomial division — example

Problem

Integrate $\int \frac{x^2+1}{x-1} dx$.

Worked solution

Divide: $(x^2 + 1)/(x - 1) = x + 1 + 2/(x - 1)$. Therefore the integral is $x^2/2 + x + 2 \log |x - 1| + C$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not start partial fractions on an improper rational function.
- Preserve excluded denominator zeros.
- Verify division by multiplying quotient and divisor and adding remainder.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Divide $(x^3 + 1)/(x + 1)$.
- 2 Integrate $(x^2 + 2x + 3)/(x + 1)$.
- 3 State the natural domain of the latter integrand.

Concise solutions

- ① $x^2 - x + 1$.
- ② Division gives $x + 1 + 2/(x + 1)$, so $x^2/2 + x + 2 \log |x + 1| + C$.
- ③ $\mathbb{R} \setminus \{-1\}$.

Partial fractions with distinct and repeated linear factors — overview

Concept

Once a rational function is proper, factor the denominator and decompose the expression into simpler fractions that can be integrated term by term.

- Distinct factors: $\frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$.
- Repeated factors require additional powers.
- Integrating then produces logarithms or reciprocal-power terms.

$$\frac{1}{(x-1)(x+2)}$$

$$= \frac{A}{x-1} + \frac{B}{x+2}$$

$$A = \frac{1}{3}, B = -\frac{1}{3}$$

$$\int \cdots dx = \frac{1}{3} \log |x-1| - \frac{1}{3} \log |x+2| + C$$

Partial fractions with distinct and repeated linear factors — example

Problem

Integrate $\int \frac{1}{x^2-1} dx$.

Worked solution

$1/(x^2 - 1) = \frac{1}{2}/(x - 1) - \frac{1}{2}/(x + 1)$. Thus $\frac{1}{2} \log |x - 1| - \frac{1}{2} \log |x + 1| + C$.

Partial fractions with distinct and repeated linear factors — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Every repeated power needs its own numerator constant.
- Factor the denominator before constructing the decomposition.
- Log absolute values arise from simple linear factors.

Partial fractions with distinct and repeated linear factors — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Decompose $1/[x(x+2)]$.
- 2 Set up the decomposition of $(2x+1)/[(x-1)^2(x+3)]$.
- 3 Integrate $1/[x(x+1)]$.

Partial fractions with distinct and repeated linear factors — solutions

Concise solutions

- ① $\frac{1}{2}(1/x) - \frac{1}{2}[1/(x+2)].$
- ② $A/(x-1) + B/(x-1)^2 + C/(x+3).$
- ③ $\log|x| - \log|x+1| + C.$

Irreducible quadratic factors and completing the square — overview

Concept

Real partial fractions for irreducible quadratics use linear numerators. Completing the square then reduces terms to logarithmic and arctangent primitives.

- For $x^2 + bx + c$ with negative discriminant, write $(x + h)^2 + a^2$.
- $\int dx/(x^2 + a^2) = \frac{1}{a} \arctan(x/a) + C$.
- A numerator proportional to the denominator derivative produces a logarithm.

Irreducible quadratic factors and completing the square — example

Problem

Integrate $\int \frac{x+1}{x^2+2x+5} dx$.

Worked solution

The denominator is $(x+1)^2 + 4$ and its derivative is $2(x+1)$. Hence the integral is $\frac{1}{2} \log(x^2 + 2x + 5) + C$.

Irreducible quadratic factors and completing the square — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Not every quadratic factors over $\mathbb{R}$.
- Check whether the numerator matches the derivative before invoking arctangent.
- Positive irreducible quadratics do not need absolute values inside the logarithm.

Irreducible quadratic factors and completing the square — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Integrate $1/(x^2 + 9)$.
- 2 Complete the square for $x^2 - 4x + 8$.
- 3 Integrate $(2x - 4)/(x^2 - 4x + 8)$.

Irreducible quadratic factors and completing the square — solutions

Concise solutions

- ① $\frac{1}{3} \arctan(x/3) + C.$
- ② $(x - 2)^2 + 4.$
- ③ $\log(x^2 - 4x + 8) + C.$

Concept

Products and powers of sine and cosine are simplified using parity and Pythagorean identities before substitution.

- If a sine power is odd, save one $\sin x$ and convert the rest using $1 - \cos^2 x$.
- If a cosine power is odd, save one $\cos x$.
- If both powers are even, use half-angle formulas.

Trigonometric integrals — example

Problem

Compute $\int \sin^3 x \cos x \, dx$.

Worked solution

Set $u = \sin x$, $du = \cos x \, dx$: the integral is $\int u^3 \, du = u^4/4 + C = \sin^4 x/4 + C$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Choose identities based on parity, not at random.
- After substitution, eliminate the other trigonometric function.
- Half-angle formulas are useful when no odd factor is available.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Integrate $\sin x \cos^2 x$.
- 2 Rewrite $\sin^2 x$ using a half-angle formula.
- 3 Integrate $\cos^2 x$.

Concise solutions

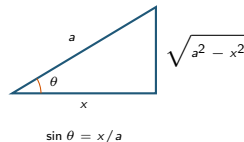
- ① $-\cos^3 x/3 + C.$
- ② $\sin^2 x = (1 - \cos 2x)/2.$
- ③ $x/2 + \sin 2x/4 + C.$

Trigonometric substitution — overview

Concept

Trigonometric substitution simplifies square roots such as $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$ by choosing x as a trigonometric function of a new angle.

- $x = a \sin \theta$ for $\sqrt{a^2 - x^2}$.
- $x = a \tan \theta$ for $\sqrt{a^2 + x^2}$.
- A right triangle helps convert back to x .



Trigonometric substitution — example

Problem

Outline the substitution for $\int dx/\sqrt{9-x^2}$.

Worked solution

Set $x = 3 \sin \theta$, so $dx = 3 \cos \theta d\theta$ and $\sqrt{9-x^2} = 3 \cos \theta$ on the chosen interval. The integral becomes $\int d\theta = \theta + C = \arcsin(x/3) + C$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Choose a domain so square-root signs are controlled.
- Back-substitute using inverse trigonometric functions or a triangle.
- Try simpler substitutions before trigonometric substitution.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Which substitution fits $\sqrt{4 + x^2}$?
- 2 Which substitution fits $\sqrt{x^2 - 25}$?
- 3 Differentiate $\arcsin(x/3)$ to verify the example.

Trigonometric substitution — solutions

Concise solutions

① $x = 2 \tan \theta.$

② $x = 5 \sec \theta.$

③ $1/\sqrt{9 - x^2}.$

Reduction formulas and integration by recurrence — overview

Concept

A reduction formula expresses an integral indexed by n in terms of one with a smaller index. This turns an infinite family into a recursive calculation.

- For $I_n = \int \sin^n x \, dx$, integration by parts gives $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$.
- Base cases are $I_0 = x$ and $I_1 = -\cos x$.
- Reduction formulas should decrease complexity and terminate.

Reduction formulas and integration by recurrence — example

Problem

Use the recurrence to compute $I_2 = \int \sin^2 x \, dx$.

Worked solution

$$I_2 = -\frac{1}{2} \sin x \cos x + \frac{1}{2} I_0 = -\frac{1}{2} \sin x \cos x + \frac{x}{2} + C, \text{ equivalent to } x/2 - \sin 2x/4 + C.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- A recurrence is incomplete without base cases.
- Track whether the index falls by 1 or 2.
- Constants of integration need not be recursively propagated at each intermediate step.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Use the recurrence to express I_4 in terms of I_2 .
- 2 Find a reduction pattern for $\int x^n e^x dx$ by integration by parts.
- 3 Why does the sine-power recurrence terminate for integer $n \geq 0$?

Concise solutions

- ① $I_4 = -\frac{1}{4} \sin^3 x \cos x + \frac{3}{4} I_2.$
- ② $J_n = x^n e^x - n J_{n-1}$, where $J_n = \int x^n e^x dx.$
- ③ Each step lowers n by 2 until reaching 0 or 1.

Concept

Improper integrals arise from infinite intervals or unbounded integrands. They are defined through limits and may converge or diverge.

- $\int_a^\infty f = \lim_{b \rightarrow \infty} \int_a^b f$ when the limit exists.
- An interior singularity requires splitting the integral.
- $\int_1^\infty x^{-p} dx$ converges iff $p > 1$.

Improper integrals and convergence — example

Problem

Determine whether $\int_1^{\infty} x^{-2} dx$ converges.

Worked solution

$\int_1^b x^{-2} dx = [-1/x]_1^b = 1 - 1/b \rightarrow 1$. It converges to 1.

Improper integrals and convergence — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Never write an infinite endpoint directly into an antiderivative.
- Each singular endpoint needs its own limit.
- Finite area is a convergence question, not merely a graphing question.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Test $\int_1^\infty dx/x$.
- 2 Compute $\int_0^1 x^{-1/2} dx$.
- 3 Why are improper integrals essential for probability densities on $\mathbb{R}$?

Concise solutions

- ① It diverges logarithmically.
- ② 2.
- ③ Total probability often requires integrating a density over an unbounded support.

Chapter 8

Linear algebra

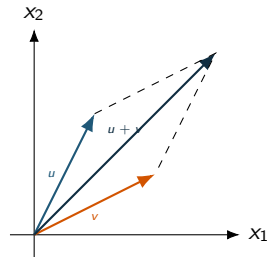
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Vectors in $\mathbb{R}^n$ and vector operations — overview

Concept

A vector in $\mathbb{R}^n$ is an ordered n -tuple. Addition and scalar multiplication are componentwise and satisfy the vector-space axioms.

- $x + y = (x_1 + y_1, \dots, x_n + y_n)$.
- $\lambda x = (\lambda x_1, \dots, \lambda x_n)$.
- The zero vector is $(0, \dots, 0)$.



Vectors in $\mathbb{R}^n$ and vector operations — example

Problem

Let $x = (1, -2, 3)$ and $y = (4, 0, -1)$. Compute $2x - y$.

Worked solution

$$2x - y = (2, -4, 6) - (4, 0, -1) = (-2, -4, 7).$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

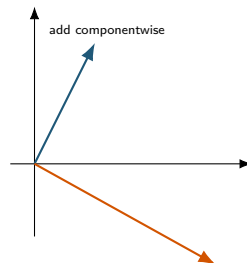
- Vector operations require matching dimensions.
- Order of coordinates matters.
- Distinguish the scalar 0 from the zero vector when context matters.

Vectors in $\mathbb{R}^n$ and vector operations — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $(1, 2) + (3, -5)$.
- 2 Find λ such that $\lambda(2, -1) = (6, -3)$.
- 3 Is $(1, 2) + (3, 4, 5)$ defined?



Concise solutions

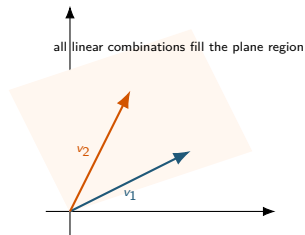
- ① $(4, -3)$.
- ② $\lambda = 3$.
- ③ No; the dimensions differ.

Linear combinations, span and linear independence — overview

Concept

A linear combination of $v_1, \dots, v_k$ is $\sum_i a_i v_i$. Their span is the set of all such combinations. Independence means that the zero vector has only the trivial representation.

- $a_1 v_1 + \dots + a_k v_k = 0$ implies all $a_i = 0$ for independence.
- A dependent family contains redundancy.
- A zero vector in a family makes the family automatically dependent.



Linear combinations, span and linear independence — example

Problem

Are $(1, 0)$ and $(2, 0)$ linearly independent?

Worked solution

No. $2(1, 0) - 1(2, 0) = (0, 0)$ is a nontrivial relation, so the vectors are dependent.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- One nontrivial zero combination is enough to prove dependence.
- Do not confuse orthogonality with independence, although nonzero orthogonal vectors are independent.
- A family containing 0 is always dependent.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Are $(1, 0)$ and $(0, 1)$ independent?
- 2 Is $\{(1, 2), (2, 4), (0, 1)\}$ independent?
- 3 Explain why $\{v, 0\}$ is dependent for every v .

Concise solutions

- ① Yes.
- ② No, because $(2, 4) = 2(1, 2)$.
- ③ $0 \cdot v + 1 \cdot 0 = 0$ is a nontrivial relation.

Concept

An $m \times n$ matrix has m rows and n columns. Matrices represent linear maps, systems, data tables and covariance structures.

- Entry notation: $A = (a_{ij})$.
- Two matrices can be added only when their dimensions match.
- The identity matrix I_n has ones on the diagonal and zeros elsewhere.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

Visual cue

This matrix has 2 rows and 3 columns, so it is a 2×3 matrix.

Matrices and dimensions — example

Problem

State the dimensions of $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ and of $A^\top$.

Worked solution

A is 2×3 ; transposition interchanges rows and columns, so $A^\top$ is 3×2 .

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Rows come first in the dimension notation.
- A matrix is not a scalar unless its dimension is 1×1 and context permits identification.
- Transpose changes dimensions.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 What is the size of a covariance matrix for 5 assets?
- 2 Can a 2×3 matrix be added to a 3×2 matrix?
- 3 Write I_3 .

Concise solutions

① 5×5 .

② No.

③ $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Matrix multiplication: row by column — overview

Concept

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$ and $(AB)_{ij}$ is row i of A dotted with column j of B .

- The inner dimensions must match.
- $(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$.
- Matrix multiplication is generally not commutative.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

$$(AB)_{12} = a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32}$$

Visual cue

Select one row from A and one column from B ; multiply matching entries and add them.

Matrix multiplication: row by column — example

Problem

Compute $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix}$.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$$

Worked solution

First row by column: $1 \cdot 5 + 2 \cdot 6 = 17$.

Second row by column: $3 \cdot 5 + 4 \cdot 6 = 39$.

Therefore

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix}.$$

Matrix multiplication: row by column — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not multiply entries componentwise.
- Write dimensions before multiplying.
- AB may exist while BA does not.

Matrix multiplication: row by column — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$.
- 2 If A is 4×2 and B is 2×7 , what is the size of AB ?
- 3 If A is 2×3 and B is 2×3 , is AB defined?

Matrix multiplication: row by column — solutions

Concise solutions

① $\begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$

② $4 \times 7.$

③ No.

Concept

The Euclidean dot product $x^\top y = \sum_i x_i y_i$ measures alignment. It induces the norm $\|x\|_2 = \sqrt{x^\top x}$ and orthogonality condition $x^\top y = 0$.

- $\|x\|_2 \geq 0$ and equals 0 only for $x = 0$.
- Cauchy–Schwarz: $|x^\top y| \leq \|x\| \|y\|$.
- Orthogonal nonzero vectors are linearly independent.

Transpose, dot product, norm and orthogonality — example

Problem

For $x = (1, 2, -1)$ and $y = (2, 0, 2)$ compute $x^\top y$.

Worked solution

$x^\top y = 1 \cdot 2 + 2 \cdot 0 + (-1) \cdot 2 = 0$, so the vectors are orthogonal.

Transpose, dot product, norm and orthogonality — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Dot products require equal vector dimensions.
- Orthogonality is stronger than zero covariance unless a probabilistic structure is specified.
- Norm is a scalar, not a vector.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $\|(3, 4)\|_2$.
- 2 Are $(1, 1)$ and $(1, -1)$ orthogonal?
- 3 Compute $(2, -1)^\top (3, 5)$.

Transpose, dot product, norm and orthogonality — solutions

Concise solutions

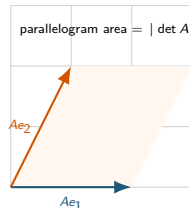
- ① 5.
- ② Yes.
- ③ 1.

Determinants and invertibility — overview

Concept

For a square matrix, the determinant measures oriented area or volume scaling. A matrix is invertible exactly when its determinant is nonzero.

- For 2×2 , $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.
- $\det A \neq 0$ means columns are independent.
- $\det A = 0$ means the transformation collapses area.



Determinants and invertibility — example

Problem

Is $A = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$ invertible?

Worked solution

$\det A = 2 \cdot 2 - 1 \cdot 4 = 0$, so A is singular and not invertible.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Determinants are defined only for square matrices.
- A small determinant numerically is not the same as an exactly zero determinant theoretically.
- Do not confuse determinant with trace.

Determinants and invertibility — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute $\det \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$.
- 2 What is $\det I_n$?
- 3 If two rows are equal, what is the determinant?

Concise solutions

- ① 13.
- ② 1.
- ③ 0.

Inverse matrices and solving $Ax=b$ — overview

Concept

For an invertible square matrix A , the inverse A^{-1} satisfies $A^{-1}A = AA^{-1} = I$. Then $Ax = b$ has the unique solution $x = A^{-1}b$.

- For 2×2 , $A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- An inverse exists only when $\det A \neq 0$.
- In computation, elimination is often preferable to explicitly forming A^{-1} .

Inverse matrices and solving $Ax=b$ — example

Problem

Solve $\begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} x = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$.

Worked solution

The inverse is $\text{diag}(1/2, 1/4)$, hence $x = (3, 2)^\top$.

Inverse matrices and solving $Ax=b$ — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Never divide by a matrix entrywise to form an inverse.
- $A^{-1}b$ requires A to be square and invertible.
- Verify a proposed solution by substitution.

Inverse matrices and solving $Ax=b$ — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Invert $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.
- 2 Solve $x + y = 3$, $x - y = 1$.
- 3 Can a singular matrix have an inverse?

Inverse matrices and solving $Ax=b$ — solutions

Concise solutions

- ① $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$.
- ② $x = 2, y = 1$.
- ③ No.

Concept

Elementary row operations transform a system into echelon form without changing its solution set. Rank counts pivot positions and controls existence and uniqueness.

- Allowed operations: swap rows, multiply a row by a nonzero scalar, add a multiple of one row to another.
- A pivot in every unknown column gives uniqueness when the system is consistent.
- Rouché–Capelli: $Ax = b$ is consistent iff $\text{rank } A = \text{rank}[A|b]$.

Gaussian elimination, rank and linear systems — example

Problem

Solve $x + y = 2$, $2x + 2y = 4$ by elimination.

Worked solution

The second row is twice the first, leaving one equation in two unknowns. There are infinitely many solutions: $x = 2 - y$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- A zero row does not automatically mean inconsistency.
- A row $0 = \text{nonzero}$ signals inconsistency.
- Rank is not simply the number of nonzero entries.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 What is the rank of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$?
- 2 What solution type occurs for $x + y = 1$, $x + y = 2$?
- 3 Solve $x + y = 3$, $2x - y = 0$.

Concise solutions

- ① 1.
- ② No solution.
- ③ $x = 1, y = 2$.

Concept

Cramer's rule expresses the unique solution of a square nonsingular system using determinants. It is mainly conceptual for larger systems but clarifies the role of $\det A$.

- For $Ax = b$, $x_i = \det A_i / \det A$.
- A_i replaces column i of A by b .
- The rule requires $\det A \neq 0$.

Cramer's rule and determinant criteria — example

Problem

Use Cramer's rule for $2x + y = 5$, $x - y = 1$.

Worked solution

$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ has determinant -3 . Replacing columns gives $\det A_1 = -6$, $\det A_2 = -3$, hence $x = 2$, $y = 1$.

Cramer's rule and determinant criteria — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not apply Cramer's rule to rectangular systems.
- Column replacement must match the unknown index.
- For numerical work, elimination is usually more stable and efficient.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 What condition is necessary for Cramer's rule?
- 2 Solve $x + y = 4$, $x - y = 2$.
- 3 Why does $\det A = 0$ prevent the Cramer formulas?

Concise solutions

- ① $\det A \neq 0$.
- ② $x = 3, y = 1$.
- ③ The denominators would vanish and uniqueness is lost.

Concept

A quadratic form has the structure $x^\top Ax$. In finance, portfolio variance is $w^\top \Sigma w$, where Σ is the covariance matrix.

- If A is symmetric positive definite, $x^\top Ax > 0$ for every $x \neq 0$.
- Covariance matrices are symmetric and positive semidefinite.
- Diagonal entries of Σ are variances.

Quadratic forms and covariance matrices — example

Problem

For $\Sigma = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}$ and $w = (1/2, 1/2)^\top$, compute $w^\top \Sigma w$.

Worked solution

$\Sigma w = (2.5, 5)^\top$, so $w^\top \Sigma w = 0.5(2.5) + 0.5(5) = 3.75$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not compute portfolio variance as a simple weighted average of individual variances.
- Covariances contribute cross terms.
- Symmetry means $\sigma_{ij} = \sigma_{ji}$.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Write the quadratic form of $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$.
- 2 What are the diagonal entries of a covariance matrix?
- 3 If two asset returns are uncorrelated, what happens to their covariance term?

Concise solutions

- ① $ax_1^2 + 2cx_1x_2 + bx_2^2$.
- ② The individual variances.
- ③ It is zero.

Chapter 9

Functions of several variables

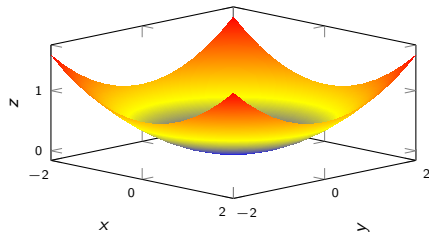
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Functions of two and several variables — overview

Concept

A function of several variables maps vectors to numbers or vectors. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the input is $x = (x_1, \dots, x_n)$ and the output is a scalar.

- The domain is now a subset of $\mathbb{R}^n$.
- A two-variable graph $z = f(x, y)$ is a surface in $\mathbb{R}^3$.
- Financial objectives often depend on many portfolio weights simultaneously.



Functions of two and several variables — example

Problem

Find the domain of $f(x, y) = \sqrt{1 - x^2 - y^2}$.

Worked solution

The radicand must be nonnegative: $x^2 + y^2 \leq 1$. The domain is the closed unit disk.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Domain restrictions become geometric regions.
- Do not treat x and y as if one were a fixed parameter unless explicitly doing a slice.
- A graph in two variables lives in three dimensions.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the domain of $\log(x - y)$.
- 2 What is the graph of $f(x, y) = x + y$?
- 3 Evaluate $f(1, 2)$ for $f(x, y) = x^2 + xy$.

Concise solutions

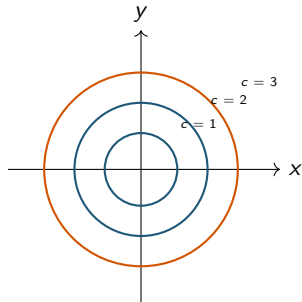
- ① $\{(x, y) : x > y\}$.
- ② A plane.
- ③ 3.

Level curves and contour interpretation — overview

Concept

A level set of f is the set of inputs producing a fixed value: $f(x, y) = c$. Level curves summarize a surface in the input plane.

- Different c values give different contours.
- Contours never cross at a point for a single-valued function unless they represent the same level.
- Utility indifference curves and iso-risk curves are level sets.



Example: $f(x, y) = x^2 + y^2$

Level curves and contour interpretation — example

Problem

Describe the level curves of $f(x, y) = x^2 + y^2$.

Worked solution

$x^2 + y^2 = c$. For $c > 0$ they are circles centered at the origin with radius $\sqrt{c}$; $c = 0$ is the origin.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Not every level value is attained.
- The geometry of level sets can reveal minima.
- Do not confuse a level curve with the graph of f .

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Describe levels of $f(x, y) = x + y$.
- 2 Describe the $c = 1$ level of xy .
- 3 What does a level set represent in a portfolio risk function?

Concise solutions

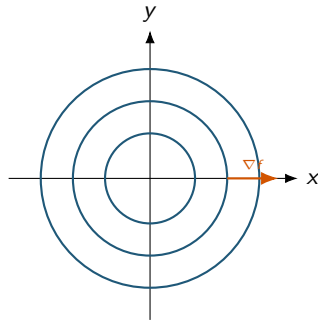
- 1 Parallel lines $x + y = c$.
- 2 The hyperbola $xy = 1$.
- 3 All portfolios with the same risk value.

Partial derivatives and gradient — overview

Concept

A partial derivative changes one input while holding the others fixed. The gradient collects all first partial derivatives.

- $f_x = \partial f / \partial x$, $f_y = \partial f / \partial y$.
- $\nabla f = (f_x, f_y)^\top$.
- The gradient points in the direction of steepest local increase.



For $f = x^2 + y^2$, the gradient is perpendicular to level curves.

Partial derivatives and gradient — example

Problem

For $f(x, y) = x^2y + e^y$, compute ∇f .

Worked solution

$f_x = 2xy$ and $f_y = x^2 + e^y$, so $\nabla f = (2xy, x^2 + e^y)^\top$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- When differentiating partially, treat the other variables as constants.
- Gradient is a vector.
- Existence of partial derivatives alone does not always guarantee differentiability.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute the gradient of $x^2 + y^2$.
- 2 Find f_x for $f(x, y) = \log(x + y)$.
- 3 Evaluate $\nabla(xy)$ at $(2, 3)$.

Concise solutions

- ① $(2x, 2y)^{\top}$.
- ② $1/(x + y)$ on $x + y > 0$.
- ③ $(3, 2)^{\top}$.

Directional derivatives and the total differential — overview

Concept

For differentiable f , the first-order change under a small vector increment h is $\nabla f(x)^\top h$. This is the multivariable analogue of linearization.

- Directional derivative along unit u : $D_u f = \nabla f^\top u$.
- $df \approx \sum_i f_{x_i} dx_i$.
- Local sensitivity is encoded by the gradient.

Directional derivatives and the total differential — example

Problem

For $f(x, y) = x^2 + y^2$, estimate the change from $(1, 2)$ under $h = (0.01, -0.02)$.

Worked solution

$\nabla f(1, 2) = (2, 4)$. Thus $\Delta f \approx (2, 4) \cdot (0.01, -0.02) = 0.02 - 0.08 = -0.06$.

Directional derivatives and the total differential — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Directional vectors are usually normalized when interpreting directional derivatives.
- Linearization is an approximation, not an exact equality except for affine functions.
- Use the gradient at the base point.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find $D_u f$ for $f = x + y$ and $u = (1/\sqrt{2}, 1/\sqrt{2})$.
- 2 Write the differential of xy .
- 3 Approximate the change in xy at $(2, 3)$ for $(dx, dy) = (0.1, 0)$.

Concise solutions

- ① $\sqrt{2}$.
- ② $d(xy) = y \, dx + x \, dy$.
- ③ 0.3.

Unconstrained extrema and the Hessian — overview

Concept

Interior differentiable extrema require a zero gradient. Second derivatives are organized in the Hessian matrix and help classify critical points.

- Critical point: $\nabla f = 0$.
- Hessian $H = (\partial^2 f / \partial x_i \partial x_j)$.
- Positive definite Hessian gives a strict local minimum; negative definite gives a strict local maximum.

Unconstrained extrema and the Hessian — example

Problem

Classify $(0, 0)$ for $f(x, y) = x^2 + 2y^2$.

Worked solution

$\nabla f = (2x, 4y)$ vanishes at $(0, 0)$. The Hessian $\text{diag}(2, 4)$ is positive definite, so the point is a strict global minimum.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- A zero gradient is necessary, not sufficient.
- Saddle points are neither maxima nor minima.
- Boundary points require separate analysis.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Classify $(0, 0)$ for $x^2 - y^2$.
- 2 Find the critical point of $(x - 1)^2 + (y + 2)^2$.
- 3 What does a positive definite Hessian indicate locally?

Unconstrained extrema and the Hessian — solutions

Concise solutions

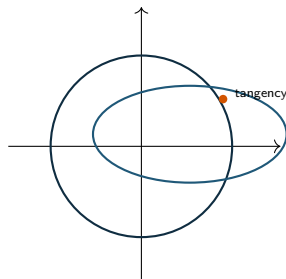
- ① Saddle point.
- ② $(1, -2)$.
- ③ Strict local convexity and a local minimum at a critical point.

Constrained optimization and Lagrange multipliers — overview

Concept

For a constrained optimization problem, the optimum often occurs where the objective and constraint have tangent level curves. This is captured by $\nabla f = \lambda \nabla g$.

- The constraint restricts feasible points.
- Tangency identifies candidate optima.
- One must also check feasibility and compare candidates.



Constrained optimization and Lagrange multipliers — example

Problem

Minimize $x^2 + y^2$ subject to $x + y = 1$.

Worked solution

$2x = \lambda$, $2y = \lambda$, so $x = y$. With $x + y = 1$, $x = y = 1/2$. The minimum is $1/2$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- The multiplier equations are candidate conditions, not automatic global conclusions.
- The constraint must be imposed alongside gradient equations.
- Degenerate constraints may require other methods.

Constrained optimization and Lagrange multipliers — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Maximize xy subject to $x + y = 10$ with $x, y \geq 0$.
- 2 Write the Lagrangian for minimizing $w^\top \Sigma w$ subject to $\mathbf{1}^\top w = 1$.
- 3 What geometric condition does $\nabla f = \lambda \nabla g$ express?

Constrained optimization and Lagrange multipliers — solutions

Concise solutions

- ① $x = y = 5$, maximum 25.
- ② $\mathcal{L} = w^\top \Sigma w - \lambda(\mathbf{1}^\top w - 1)$.
- ③ At a regular constrained optimum, objective and constraint gradients are parallel.

Chapter 10

Descriptive statistics

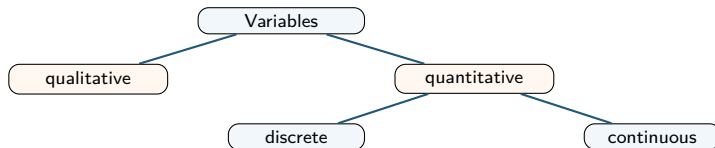
7 detailed review units $\times$ five frames per unit

Statistical units, variables and measurement scales — overview

Concept

Statistics begins by identifying the population or collection of units and the variables measured on them. Variable type determines which summaries and operations are meaningful.

- Qualitative variables describe categories; quantitative variables are numerical.
- Quantitative variables can be discrete or continuous.
- Nominal, ordinal, interval and ratio scales support progressively richer operations.



Statistical units, variables and measurement scales — example

Problem

Classify sector, credit rating, number of trades, and daily return.

Worked solution

Sector is nominal qualitative; credit rating is ordinal qualitative; number of trades is discrete quantitative; daily return is quantitative and typically treated as continuous.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Numerical labels do not automatically make a variable quantitative.
- An ordinal scale supports ranking but not meaningful arithmetic gaps.
- Always identify the statistical unit as well as the variable.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Classify a binary default indicator.
- 2 Classify annual income.
- 3 Give a financial example of an ordinal variable.

Concise solutions

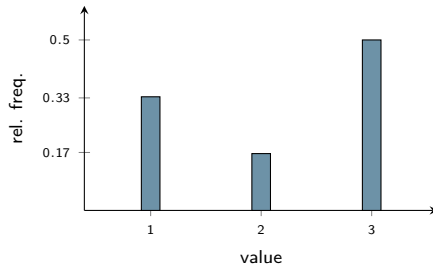
- ① Qualitative binary or equivalently Bernoulli-coded quantitative indicator, depending on purpose.
- ② Quantitative, ratio scale.
- ③ Credit rating class is a standard example.

Frequency distributions, grouped data and empirical CDF — overview

Concept

Frequency tables compress raw data into counts, proportions and cumulative proportions. Continuous data may be grouped into classes, while the empirical distribution function records cumulative relative frequency at every threshold.

- Relative frequency $f_i = n_i/n$.
- Cumulative frequencies add classes in order.
- With unequal class widths, histogram height should use frequency density.



Frequency distributions, grouped data and empirical CDF — example

Problem

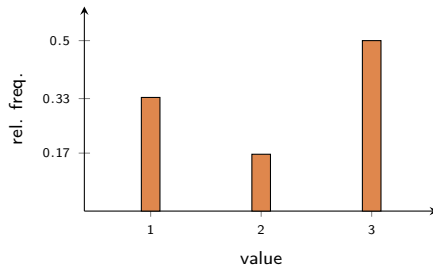
For data 1, 1, 2, 3, 3, 3, construct relative frequencies.

Worked solution

Counts are 2, 1, 3 for values 1, 2, 3, so relative frequencies are

$$\frac{2}{6} = \frac{1}{3}, \quad \frac{1}{6}, \quad \frac{3}{6} = \frac{1}{2}.$$

The bar plot gives an immediate visual summary: value 3 is the most frequent.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Histogram area, not necessarily bar height, represents frequency when widths differ.
- Cumulative frequencies require an ordered variable.
- Do not confuse a density with a probability at one exact continuous value.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute the empirical CDF at $x = 2$ for the sample above.
- 2 If a class has frequency 20 and width 4, find frequency density.
- 3 Why is a bar chart different from a histogram?

Concise solutions

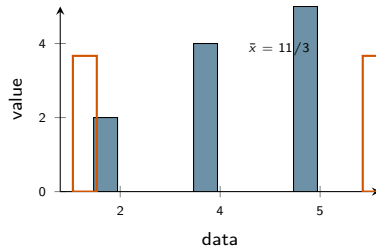
- ① $F_n(2) = 3/6 = 1/2$.
- ② 5.
- ③ Bars represent categories/discrete values; histogram intervals form a numerical scale and area carries frequency.

Means: arithmetic, weighted, geometric and quadratic — overview

Concept

Different means answer different questions. Arithmetic means average additive quantities, weighted means reflect unequal importance, and geometric means summarize multiplicative growth.

- $\bar{x} = n^{-1} \sum_i x_i$.
- $\bar{x}_w = \sum_i w_i x_i$ with $\sum_i w_i = 1$.
- $G = (\prod_i x_i)^{1/n}$ for positive data.
- Quadratic mean $Q = \sqrt{n^{-1} \sum_i x_i^2}$.



Means: arithmetic, weighted, geometric and quadratic — example

Problem

A portfolio has weights $(0.4, 0.6)$ and returns $(5\%, -2\%)$. Find portfolio return.

Worked solution

$$R_p = 0.4(0.05) + 0.6(-0.02) = 0.008 = 0.8\%.$$

Means: arithmetic, weighted, geometric and quadratic — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Use geometric, not arithmetic, averaging for compounded growth factors over time.
- Weights need not be equal but should match the intended normalization.
- Percentage returns should be converted consistently to decimals during calculation.

Means: arithmetic, weighted, geometric and quadratic — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the arithmetic mean of 2, 4, 9.
- 2 Find the weighted mean of 10, 20 with weights 0.25, 0.75.
- 3 Find the geometric mean of growth factors 1.10 and 0.90.

Means: arithmetic, weighted, geometric and quadratic — solutions

Concise solutions

- ① 5.
- ② 17.5.
- ③ $\sqrt{0.99} \approx 0.995$.

Concept

Quantiles describe positions in an ordered distribution. The median is the 0.5 quantile and is typically more robust to extreme values than the arithmetic mean.

- Quartiles correspond approximately to 25%, 50%, 75% cumulative positions.
- The mode is the most frequent value or class.
- Quantile conventions differ slightly in finite samples, so software definitions should be documented.

Median, quantiles, mode and robust location — example

Problem

Compare mean and median for 1, 2, 2, 3, 100.

Worked solution

Mean = 21.6, while median = 2. The extreme value strongly affects the mean but not the median.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Quantiles require ordered data.
- Different software may interpolate quantiles differently.
- Robust does not mean statistically optimal in every model.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the median of 1, 4, 7, 10 using midpoint convention.
- 2 Find the mode of 1, 1, 2, 3, 3, 3.
- 3 Which location measure is less sensitive to a single extreme outlier?

Concise solutions

- ① 5.5.
- ② 3.
- ③ The median.

Dispersion: range, IQR, variance and standard deviation — overview

Concept

Dispersion measures quantify how far observations spread around a center. Variance squares deviations; standard deviation restores the original measurement unit.

- Range = $\max x_i - \min x_i$.
- IQR = $Q_3 - Q_1$.
- Descriptive variance can use $n^{-1} \sum (x_i - \bar{x})^2$.
- In inference, the unbiased sample variance uses denominator $n - 1$.

Dispersion: range, IQR, variance and standard deviation — example

Problem

For data 1, 2, 3, compute the descriptive variance.

Worked solution

$\bar{x} = 2$. Squared deviations are 1, 0, 1, so variance = $2/3$ and standard deviation = $\sqrt{2/3}$.

Dispersion: range, IQR, variance and standard deviation — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Variance is in squared units.
- State whether denominator n or $n - 1$ is being used.
- A small variance does not mean a small mean.

Dispersion: range, IQR, variance and standard deviation — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find the range of $-2, 4, 10$.
- 2 Show that adding a constant to every observation leaves variance unchanged.
- 3 What happens to standard deviation when all data are multiplied by 3?

Dispersion: range, IQR, variance and standard deviation — solutions

Concise solutions

- ① 12.
- ② All deviations from the shifted mean remain the same.
- ③ It is multiplied by 3 in absolute value.

Concept

Location and dispersion do not fully describe distribution shape. Skewness summarizes asymmetry, kurtosis relates to tail/peak behavior, and boxplots provide a robust graphical summary. Concentration measures address how a total is distributed across units.

- Positive skewness is associated with a longer right tail.
- Boxplots display quartiles, median and an outlier rule.
- Lorenz curves and Gini-type indices summarize concentration.
- Financial return distributions often motivate attention to asymmetry and tails.

Shape, skewness, kurtosis, boxplots and concentration — example

Problem

Two samples have the same mean and variance, but one contains more extreme tail observations. What statistic or graph could reveal the difference?

Worked solution

Kurtosis-related measures and a boxplot can reveal tail/outlier differences; quantile plots can also help.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not equate kurtosis solely with “peakedness”.
- Outlier rules are conventions, not proofs that observations are errors.
- Skewness near zero does not imply normality.

Shape, skewness, kurtosis, boxplots and concentration — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- ① What does a right-skewed distribution look like?
- ② What does IQR measure in a boxplot?
- ③ What does perfect concentration mean conceptually?

Concise solutions

- ① A longer or heavier right tail.
- ② The width of the middle 50% of observations.
- ③ One unit holds the entire total amount.

Concept

Joint data permit analysis of dependence. Covariance measures signed co-movement, correlation standardizes it, and simple least-squares regression summarizes a linear conditional pattern descriptively.

- s_{XY} averages products of centered observations under the chosen normalization.
- $r = s_{XY}/(s_X s_Y) \in [-1, 1]$.
- Zero correlation does not generally imply independence.
- OLS minimizes $\sum_i (y_i - b_0 - b_1 x_i)^2$.

Bivariate distributions, covariance, correlation and descriptive regression — example

Problem

If $Y = 2X + 1$ exactly and X varies, what is the correlation?

Worked solution

The relation is perfectly increasing and affine, so correlation is $+1$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Correlation measures linear association, not causality.
- Regression residuals are vertical deviations from fitted values.
- A strong fitted relationship may be driven by outliers.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 What is correlation if $Y = -3X + 5$ exactly and X varies?
- 2 Can $r = 0$ occur for a strong nonlinear relation?
- 3 What quantity does least squares minimize?

Concise solutions

- ① -1 .
- ② Yes.
- ③ The sum of squared residuals.

Chapter 11

Probability

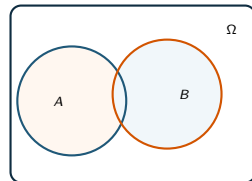
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Random experiments, sample spaces and events — overview

Concept

A random experiment has an outcome that is uncertain before observation. The sample space Ω collects all possible outcomes, and events are subsets of Ω .

- Elementary outcomes are points $\omega \in \Omega$.
- An event occurs when the realized outcome belongs to its subset.
- Set operations become logical operations on events.



Random experiments, sample spaces and events — example

Problem

For two coin tosses, define Ω and the event $A = \text{"exactly one head"}$.

Worked solution

$\Omega = \{HH, HT, TH, TT\}$ and $A = \{HT, TH\}$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Define the experiment before assigning probabilities.
- Outcomes must be mutually distinguishable and collectively exhaustive.
- Events need not be single outcomes.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 For a die roll, define the event “even”.
- 2 For two coin tosses, define “at least one head”.
- 3 What is the impossible event?

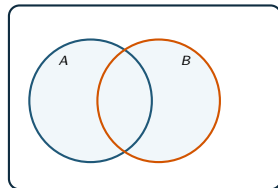
Concise solutions

- ① $\{2, 4, 6\}$.
- ② $\{HH, HT, TH\}$.
- ③ $\emptyset$.

Concept

A probability measure assigns values in $[0, 1]$ to events and is normalized and countably additive on disjoint events.

- $P(\Omega) = 1$, $P(\emptyset) = 0$.
- $P(A^c) = 1 - P(A)$.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- If $A \subseteq B$, then $P(A) \leq P(B)$.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Probability axioms and elementary rules — example

Problem

If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find $P(A \cup B)$.

Worked solution

$$P(A \cup B) = 0.6 + 0.5 - 0.3 = 0.8.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not add probabilities of overlapping events without subtracting overlap.
- Probability zero does not necessarily mean logically impossible in continuous models.
- All probabilities must lie between 0 and 1.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- ① If $P(A) = 0.2$, find $P(A^c)$.
- ② If A and B are disjoint with probabilities 0.3 and 0.4, find $P(A \cup B)$.
- ③ Can $P(A) = 1.2$?

Concise solutions

- ① 0.8.
- ② 0.7.
- ③ No.

Concept

In finite equally likely models, counting often converts event size into probability. Permutations count ordered selections; combinations count unordered selections.

- $n! = n(n-1) \cdots 1$.
- Ordered selection without replacement: $n!/(n-k)!$.
- Unordered selection: $\binom{n}{k} = n!/[k!(n-k)!]$.

Counting, factorials, permutations and combinations — example

Problem

How many unordered committees of 3 can be chosen from 8 people?

Worked solution

$$\binom{8}{3} = 8!/(3!5!) = 56.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Decide whether order matters before choosing a formula.
- Replacement changes the counting problem.
- Equally likely outcomes are required for the ratio $\#A/\#\Omega$.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 How many ordered pairs of distinct people can be chosen from 8?
- 2 Compute $\binom{5}{2}$.
- 3 How many binary strings of length 4 exist?

Concise solutions

① $8 \cdot 7 = 56.$

② $10.$

③ $2^4 = 16.$

Concept

Conditional probability rescales probabilities after learning that an event has occurred. It is the foundation of sequential reasoning.

- $P(A \mid B) = P(A \cap B)/P(B)$ for $P(B) > 0$.
- $P(A \cap B) = P(A \mid B)P(B)$.
- Tree diagrams help organize repeated conditioning.

Conditional probability and multiplication rule — example

Problem

Suppose $P(B) = 0.4$ and $P(A \cap B) = 0.1$. Find $P(A \mid B)$.

Worked solution

$$P(A \mid B) = 0.1/0.4 = 0.25.$$

Conditional probability and multiplication rule — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Conditioning changes the reference population.
- $P(A \mid B)$ and $P(B \mid A)$ are generally different.
- Never condition on an event of probability zero using the elementary formula.

Conditional probability and multiplication rule — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $P(A | B) = 0.7$ and $P(B) = 0.2$, find $P(A \cap B)$.
- 2 Interpret $P(\text{default} | \text{low rating})$.
- 3 Is conditional probability symmetric?

Concise solutions

- ① 0.14.
- ② Default probability within the subpopulation having low rating.
- ③ No.

Concept

Events are independent when knowing one does not change the probability of the other. For independent repeated trials, joint probabilities factor.

- $P(A \cap B) = P(A)P(B)$.
- If $P(B) > 0$, independence implies $P(A \mid B) = P(A)$.
- Pairwise independence and mutual independence are distinct for more than two events.

Independence and repeated trials — example

Problem

Two independent default events have probabilities 0.02 and 0.03. Find the probability both occur.

Worked solution

$$0.02 \cdot 0.03 = 0.0006.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Disjoint nonzero-probability events are not independent.
- Uncorrelated random variables need not be independent.
- Independence is a model assumption that should be justified.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $P(A) = 0.5$, $P(B) = 0.4$, independent, find $P(A \cup B)$.
- 2 Can mutually exclusive events with positive probabilities be independent?
- 3 For three independent fair coin tosses, find $P(HHH)$.

Independence and repeated trials — solutions

Concise solutions

- ① $0.5 + 0.4 - 0.2 = 0.7$.
- ② No.
- ③ $1/8$.

Concept

A partition decomposes an event into mutually exclusive scenarios. The law of total probability aggregates scenario probabilities, and Bayes' theorem reverses conditioning.

- $P(B) = \sum_i P(B \mid A_i)P(A_i)$ for a partition (A_i) .
- $P(A_i \mid B) = P(B \mid A_i)P(A_i)/P(B)$.
- Bayesian updating combines prior probabilities with likelihood information.

Total probability and Bayes' theorem — example

Problem

A test is positive with probability 0.9 under condition D and 0.1 without it. If $P(D) = 0.05$, find $P(D \mid +)$.

Worked solution

$P(+) = 0.9(0.05) + 0.1(0.95) = 0.14$. Thus $P(D \mid +) = 0.045/0.14 \approx 0.3214$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not confuse sensitivity with posterior probability.
- Base rates matter strongly.
- The partition probabilities must sum to one.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $P(A) = 0.4$, $P(B | A) = 0.8$, $P(B | A^c) = 0.2$, find $P(B)$.
- 2 Then find $P(A | B)$.
- 3 Why is Bayes useful in credit-risk classification?

Concise solutions

- ① $0.8(0.4) + 0.2(0.6) = 0.44.$
- ② $0.32/0.44 \approx 0.7273.$
- ③ It updates default/class probabilities after observing new signals.

Concept

Probability can play different roles. Frequentist probabilities describe long-run frequencies, subjective probabilities encode beliefs, while risk-neutral probabilities are mathematical pricing measures chosen to value discounted payoffs consistently.

- Physical probability concerns real-world event likelihood.
- Risk-neutral probability is not generally a forecast of actual frequencies.
- Changing probability measure can change expected returns while preserving the set of scenarios.

Physical, subjective and risk-neutral probabilities — example

Problem

A derivative is priced using a risk-neutral expected discounted payoff. Does that mean the risk-neutral probability of a crash is the market's literal forecast?

Worked solution

Not necessarily. Risk-neutral probabilities incorporate pricing and risk premia; they are pricing devices unless additional assumptions connect them to physical beliefs.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not use “probability” without specifying the model or measure when multiple measures are present.
- Pricing probabilities and historical frequencies answer different questions.
- This package introduces the distinction; rigorous measure change belongs to the course.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- ① Which probability is estimated from historical default frequencies?
- ② Which probability measure is typically used in arbitrage-free derivative pricing?
- ③ Can two measures assign different expectations to the same payoff?

Physical, subjective and risk-neutral probabilities — solutions

Concise solutions

- ① A physical/statistical probability model.
- ② A risk-neutral (equivalent martingale) measure under standard settings.
- ③ Yes.

Chapter 12

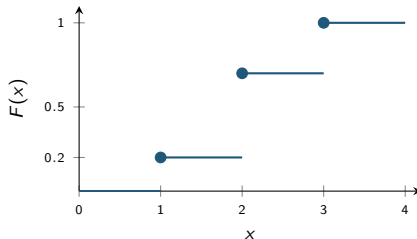
Random variables and probability models

6 detailed review units $\times$ five frames per unit

Concept

A random variable $X : \Omega \rightarrow \mathbb{R}$ converts uncertain outcomes into numbers. Discrete variables are described by probability mass functions; continuous variables by densities, while every real-valued random variable has a cumulative distribution function.

- $p_X(x) = P(X = x)$ for a discrete variable.
- $P(a \leq X \leq b) = \int_a^b f_X(x) dx$ for a continuous density.
- $F_X(x) = P(X \leq x)$ is nondecreasing and right-continuous.



Random variables, PMFs, PDFs and CDFs — example

Problem

If X takes 0, 1, 2 with probabilities 0.2, 0.5, 0.3, find $F_X(1)$.

Worked solution

$$F_X(1) = P(X \leq 1) = 0.2 + 0.5 = 0.7.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- A density value is not itself a probability at a point.
- Continuous variables satisfy $P(X = x) = 0$ for every fixed x under an ordinary density.
- The CDF contains complete distributional information.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find $F_X(0)$ for the example.
- 2 If $f(x) = 1$ on $[0, 1]$, find $P(X \leq 0.4)$.
- 3 What are the limiting values of a CDF as $x \rightarrow \pm\infty$?

Concise solutions

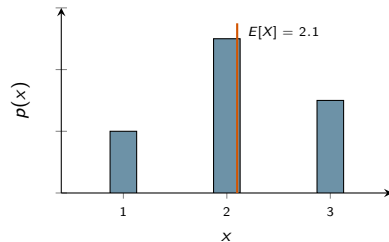
- ① 0.2.
- ② 0.4.
- ③ 0 at $-\infty$ and 1 at $+\infty$.

Expectation, moments and transformations — overview

Concept

Expectation is the probability-weighted average of a random variable. More generally, $E[g(X)]$ summarizes transformed outcomes without first deriving the full distribution of $g(X)$.

- Discrete: $E[X] = \sum_x xp_X(x)$.
- Continuous: $E[X] = \int xf_X(x) dx$ when integrable.
- Linearity: $E[aX + b] = aE[X] + b$.



Expectation, moments and transformations — example

Problem

For $P(X = 0) = 0.25$, $P(X = 2) = 0.75$, compute $E[X]$ and $E[X^2]$.

Worked solution

$E[X] = 0 + 1.5 = 1.5$. Also $E[X^2] = 0 + 4(0.75) = 3$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Expectation need not be an attainable outcome.
- Existence of a distribution does not guarantee every moment is finite.
- Linearity of expectation does not require independence.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $E[X] = 3$, find $E[2X - 5]$.
- 2 Compute $E[X]$ for a fair die.
- 3 Why is $E[X^2]$ useful for variance?

Concise solutions

- ① 1.
- ② 3.5.
- ③ $\text{Var}(X) = E[X^2] - E[X]^2$.

Concept

Variance measures squared dispersion around the mean. Covariance measures joint centered movement, and correlation standardizes covariance.

- $\text{Var}(X) = E[X^2] - E[X]^2$.
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
- $\text{Cov}(X, Y) = E[(X - EX)(Y - EY)]$.
- $Z = (X - \mu)/\sigma$ has mean 0 and variance 1 when $\sigma > 0$.

Variance, standardization, covariance and correlation — example

Problem

If $E[X] = 2$ and $E[X^2] = 7$, find $\text{Var}(X)$.

Worked solution

$$7 - 2^2 = 3.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Adding a constant changes the mean but not the variance.
- Covariance depends on scale; correlation does not.
- Zero covariance alone does not imply independence in general.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $\text{Var}(X) = 4$, find $\text{Var}(3X - 2)$.
- 2 If $\text{Cov}(X, Y) = 0$ and variables are jointly normal, what can be concluded?
- 3 What is the variance of a standardized variable?

Concise solutions

- ① 36.
- ② They are independent under joint normality.
- ③ 1.

Concept

Bernoulli models a single binary trial, binomial counts successes across independent identical Bernoulli trials, and Poisson models counts with a rate parameter.

- Bernoulli(p): $E[X] = p$, $\text{Var}(X) = p(1 - p)$.
- Binomial(n, p): $E[X] = np$, $\text{Var}(X) = np(1 - p)$.
- Poisson(λ): $E[X] = \text{Var}(X) = \lambda$.

Bernoulli, binomial and Poisson models — example

Problem

For $X \sim \text{Binomial}(10, 0.2)$, find $P(X = 0)$.

Worked solution

$$P(X = 0) = (0.8)^{10}.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Binomial requires a fixed number of trials and common success probability under the standard model.
- Poisson support is all nonnegative integers.
- Do not confuse the probability parameter p with an observed sample proportion.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Find $E[X]$ for $\text{Binomial}(20, 0.05)$.
- 2 Find $P(X = 1)$ for $\text{Poisson}(2)$.
- 3 What distribution models a default indicator?

Concise solutions

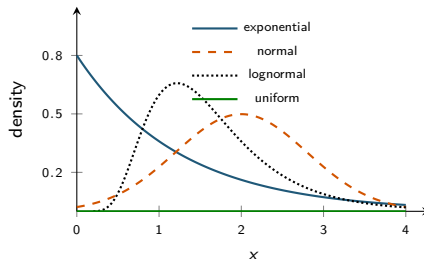
- ① 1.
- ② $2e^{-2}$.
- ③ Bernoulli.

Uniform, normal, exponential and lognormal models — overview

Concept

Continuous probability models provide reusable shapes for uncertainty. The uniform has constant density on an interval; the normal is bell-shaped; the exponential models positive waiting times; the lognormal arises when the logarithm is normal.

- $\text{Uniform}(a, b)$ density is $1/(b - a)$ on $[a, b]$.
- Normal is determined by mean μ and variance σ^2 .
- $\text{Exponential}(\lambda)$ has density $\lambda e^{-\lambda x}$ for $x \geq 0$.
- Lognormal distributions are positive and right-skewed.



Uniform, normal, exponential and lognormal models — example

Problem

If $X \sim U(2, 6)$, compute $P(3 \leq X \leq 5)$.

Worked solution

The interval length is 2 out of total length 4, so the probability is $1/2$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Normal variables can take negative values.
- Lognormal variables are strictly positive.
- Model choice should reflect support and empirical behavior, not convenience alone.

Uniform, normal, exponential and lognormal models — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Standardize $X = 12$ when $\mu = 10$, $\sigma = 2$.
- 2 Find $E[X]$ for an exponential variable with rate λ .
- 3 Why can lognormal distributions be useful for positive prices?

Uniform, normal, exponential and lognormal models — solutions

Concise solutions

- ① $Z = 1$.
- ② $1/\lambda$.
- ③ They have positive support and arise naturally from exponentiating normal log-values.

Concept

Joint distributions describe several random variables together. Their dependence determines variances of sums, while the law of large numbers and central limit theorem explain the behavior of averages.

- $E[aX + bY] = aEX + bEY$.
- $\text{Var}(aX + bY) = a^2 \text{Var } X + b^2 \text{Var } Y + 2ab \text{Cov}(X, Y)$.
- For i.i.d. finite-mean data, $\bar{X}_n \rightarrow \mu$ in probability.
- With finite variance, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \Rightarrow N(0, 1)$.

Joint distributions, sums, LLN and CLT — example

Problem

If independent X, Y have variances 4 and 9, find $\text{Var}(X + Y)$.

Worked solution

Independence implies covariance 0, so the variance is $4 + 9 = 13$.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Variances do not add without a covariance term unless covariance is zero.
- LLN concerns convergence of averages; CLT concerns the scaled distribution of their errors.
- Time-series dependence can invalidate the elementary i.i.d. formulas.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $\text{Cov}(X, Y) = 2$, find $\text{Var}(X + Y)$ using the same variances.
- 2 What happens to $\text{Var}(\bar{X}_n)$ for i.i.d. variance σ^2 ?
- 3 What theorem motivates normal approximations to sample means?

Concise solutions

- ① $4 + 9 + 2(2) = 17.$
- ② $\sigma^2/n.$
- ③ The central limit theorem.

Chapter 13

Sampling and statistical inference

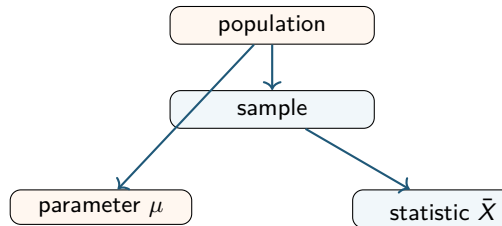
7 detailed review units $\times$ five frames per unit

Population, sample, parameter, statistic and estimator — overview

Concept

Inference separates unknown population quantities from random sample summaries. A parameter is fixed but unknown; a statistic is computed from the sample and is therefore random before observation.

- Population mean μ is a parameter; sample mean $\bar{X}$ is a statistic.
- An estimator is a statistic used to estimate a parameter.
- The observed numerical value of an estimator is an estimate.



Population, sample, parameter, statistic and estimator — example

Problem

A sample of daily returns has average 0.0012. Distinguish μ , $\bar{X}$, and 0.0012.

Worked solution

μ is the unknown population mean return; $\bar{X}$ is the random sample-mean estimator; 0.0012 is the realized estimate from this data set.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not call an observed estimate a population parameter.
- Sampling uncertainty remains even when computation is exact.
- The data-generating mechanism matters for interpretation.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Is sample variance a parameter or a statistic?
- 2 Give an estimator for a Bernoulli success probability.
- 3 Why is an estimator random before data are observed?

Concise solutions

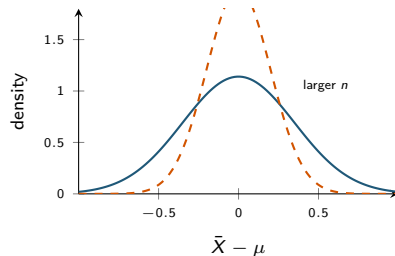
- ① A statistic.
- ② The sample proportion $\hat{p}$.
- ③ It is a function of random observations.

Sampling distributions and standard errors — overview

Concept

The sampling distribution describes how an estimator would vary across repeated samples. Its standard deviation is called the standard error.

- For i.i.d. data with variance σ^2 , $SE(\bar{X}) = \sigma/\sqrt{n}$.
- Estimated standard errors replace unknown parameters by sample estimates.
- Precision generally improves with sample size.



Sampling distributions and standard errors — example

Problem

If $\sigma = 10$ and $n = 100$, find the standard error of the sample mean.

Worked solution

$$10/\sqrt{100} = 1.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Standard deviation describes individual observations; standard error describes an estimator.
- Doubling n does not halve the standard error.
- Dependence can change standard-error formulas.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If n grows from 100 to 400, by what factor does SE fall?
- 2 If $\sigma = 6$, $n = 36$, find SE.
- 3 Can an estimator have small standard error but large bias?

Concise solutions

- ① It is halved.
- ② 1.
- ③ Yes. Precision and bias are different properties.

Bias, variance, MSE and consistency — overview

Concept

Estimator quality has several dimensions. Unbiasedness concerns expected error, variance concerns dispersion, MSE combines both, and consistency concerns large-sample convergence.

- $\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$.
- $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$.
- Consistency means $\hat{\theta}_n \rightarrow \theta$ in probability.



Intervals concentrate around θ as n grows.

Bias, variance, MSE and consistency — example

Problem

Estimator A has bias 1 and variance 4; estimator B is unbiased with variance 6. Which has smaller MSE?

Worked solution

$\text{MSE}_A = 4 + 1 = 5$; $\text{MSE}_B = 6$. A has the smaller MSE despite bias.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Unbiasedness does not guarantee minimum MSE.
- Consistency is asymptotic, not a finite-sample guarantee.
- A low MSE statement must specify the parameter/model context.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute MSE for bias -2 and variance 3 .
- 2 Can a biased estimator be consistent?
- 3 What two components form MSE?

Concise solutions

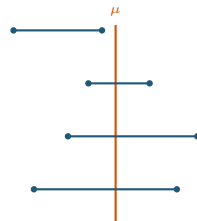
- ① 7.
- ② Yes, if the bias vanishes and variance contracts appropriately as n grows.
- ③ Variance and squared bias.

Confidence intervals and their interpretation — overview

Concept

A frequentist confidence interval is a random procedure with prescribed long-run coverage. Once the interval is observed, the parameter is fixed; the randomness belonged to the sampling procedure.

- Generic form: estimate $\pm$ critical value $\times$ standard error.
- Coverage $1 - \alpha$ refers to repeated sampling.
- Interval width reflects both uncertainty and chosen confidence level.



Most intervals capture the true parameter line; some do not.

Confidence intervals and their interpretation — example

Problem

A 95% interval for μ is (1.2, 2.8). Is it correct to say there is a 95% frequentist probability that this fixed interval contains μ ?

Worked solution

Strictly, no. The method has 95% coverage across repeated samples. After observing the interval, it either contains the fixed parameter or it does not.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not confuse confidence with Bayesian posterior probability.
- A wider confidence level usually produces a wider interval.
- A narrow interval can still be systematically biased if assumptions fail.

Confidence intervals and their interpretation — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 What happens to CI width when n increases, other things equal?
- 2 What does 99% confidence do relative to 95%?
- 3 What quantity multiplies the critical value in a standard CI?

Concise solutions

- ① It typically decreases at rate about $1/\sqrt{n}$.
- ② It uses a larger critical value and gives a wider interval.
- ③ The standard error.

Concept

Different pivots are used depending on what is known and on distributional assumptions. Normal, Student t , and chi-square distributions support standard intervals for means and variances.

- Known σ : $\bar{X} \pm z_{1-\alpha/2}\sigma/\sqrt{n}$.
- Normal population, unknown σ : $\bar{X} \pm t_{n-1,1-\alpha/2}S/\sqrt{n}$.
- For normal data, $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$.

Z, t and chi-square confidence intervals — example

Problem

Why is a t critical value larger than the corresponding normal critical value for small samples?

Worked solution

Estimating σ adds uncertainty. The Student distribution has heavier tails, so its high quantiles are larger; it approaches the normal as degrees of freedom increase.

Z, t and chi-square confidence intervals — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Use the correct degrees of freedom.
- Exact small-sample t inference relies on normality in the classical model.
- Variance intervals are asymmetric because chi-square quantiles are asymmetric.

Z, t and chi-square confidence intervals — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 What degrees of freedom are used for a one-sample t interval with $n = 20$?
- 2 When does t_{n-1} approach the standard normal?
- 3 Which distribution is used for exact normal-population variance inference?

Z, t and chi-square confidence intervals — solutions

Concise solutions

- ① 19.
- ② As n becomes large.
- ③ Chi-square.

Concept

A hypothesis test specifies a null H_0 , an alternative H_1 , a statistic, and a rejection rule. Type I error rejects a true null; Type II error fails to reject a false null.

- Significance level $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$.
- Power is $P(\text{reject } H_0 \mid H_1 \text{ true})$ for a specified alternative.
- One-sided and two-sided alternatives produce different rejection regions.

Hypothesis tests, errors, significance and power — example

Problem

A test uses $\alpha = 0.05$. Interpret the number 0.05.

Worked solution

Under the null model and repeated use of the procedure, the probability of a Type I rejection is controlled at 5%. It is not the probability that the null is true.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Failure to reject is not proof that H_0 is true.
- Power depends on effect size, sample size, noise and the test.
- Choose one- vs two-sided alternatives before looking at the data.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Define Type II error.
- 2 What usually happens to power when sample size increases?
- 3 Can lowering α while holding everything else fixed reduce power?

Concise solutions

- ① Not rejecting H_0 when the specified alternative is true.
- ② It increases.
- ③ Yes.

Concept

The p-value measures how incompatible the observed test statistic is with the null model, in the sense of tail extremeness specified by the test. It is a continuous measure of evidence, not an effect-size measure.

- Reject at level α when $p \leq \alpha$ under the standard rule.
- A p-value is computed assuming H_0 .
- Statistical significance does not imply economic or practical importance.

p-values, critical values and practical significance — example

Problem

A huge sample yields an estimated effect of 0.001 with $p = 10^{-6}$. What should be concluded?

Worked solution

There is strong statistical evidence against the exact null, but the estimated effect is tiny. Economic relevance must be assessed separately with scale, confidence intervals and context.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- p is not $P(H_0 \mid \text{data})$.
- Do not report only “significant/not significant” without effect estimates.
- Multiple testing can inflate false-positive risk.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $p = 0.03$, is the result significant at 5%? at 1%?
- 2 What happens to a two-sided p-value when the observed standardized statistic becomes more extreme?
- 3 Why report a confidence interval alongside a p-value?

Concise solutions

- ① Yes at 5%, no at 1%.
- ② It decreases.
- ③ The interval communicates effect size and precision, not merely a threshold decision.

Chapter 14

Linear regression and statistical interpretation

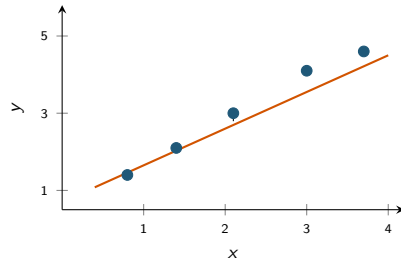
4 detailed review units $\times$ five frames per unit

Simple OLS: coefficients and geometry — overview

Concept

In the simple model $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, ordinary least squares chooses the line minimizing the sum of squared vertical residuals.

- $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$.
- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.
- With an intercept, OLS residuals sum to zero.



Simple OLS: coefficients and geometry — example

Problem

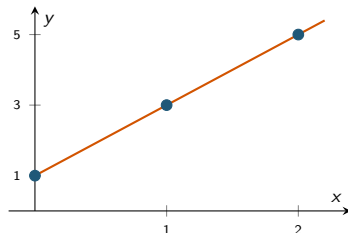
For data $(x, y) = (0, 1), (1, 3), (2, 5)$, find the fitted line.

Worked solution

The three points lie exactly on the line $y = 1 + 2x$. Hence

$$\hat{\beta}_0 = 1, \quad \hat{\beta}_1 = 2,$$

and every residual is zero.



Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- OLS slope is undefined when all x_i are identical.
- Residuals are observed-minus-fitted, not the unobserved model errors themselves.
- Association alone does not establish causation.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $\hat{\beta}_1 = 0$, what does the fitted line look like?
- 2 Write the residual e_i .
- 3 Which quantity does OLS minimize?

Simple OLS: coefficients and geometry — solutions

Concise solutions

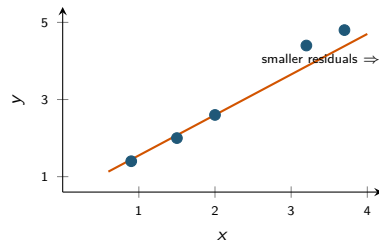
- ① Horizontal at $\hat{\beta}_0 = \bar{y}$.
- ② $e_i = y_i - \hat{y}_i$.
- ③ $\sum_i e_i^2$.

R-squared, residual scale and prediction — overview

Concept

R^2 summarizes the fraction of sample variation in Y explained by the fitted linear model with an intercept. Residual diagnostics and out-of-sample prediction are needed to assess more than in-sample fit.

- $R^2 = 1 - \text{SSE}/\text{SST}$.
- $R^2 = 1$ under perfect in-sample fit.
- A high R^2 does not guarantee causal validity or forecasting performance.



R-squared, residual scale and prediction — example

Problem

If $SST = 100$ and $SSE = 25$, compute R^2 .

Worked solution

$$R^2 = 1 - 25/100 = 0.75.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not compare R^2 blindly across different dependent variables.
- Overfitting can improve in-sample fit while harming prediction.
- Inspect residual patterns for misspecification.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Compute R^2 when SSE equals SST.
- 2 What does a residual plot with a clear curve suggest?
- 3 Can R^2 alone establish economic significance?

Concise solutions

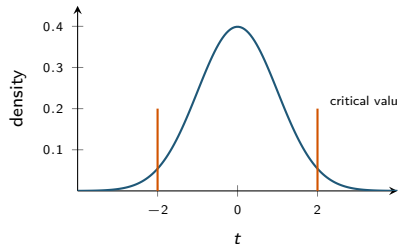
- ① 0.
- ② The linear functional form may be inadequate.
- ③ No.

Inference for the regression slope — overview

Concept

Under the classical regression assumptions, the estimated slope has a sampling distribution and a standard error. A common test is $H_0 : \beta_1 = 0$.

- $t = (\hat{\beta}_1 - \beta_{1,0}) / \text{SE}(\hat{\beta}_1)$.
- Confidence intervals combine the slope estimate and its standard error.
- The null $\beta_1 = 0$ concerns the linear coefficient within the specified model.



Inference for the regression slope — example

Problem

Suppose $\hat{\beta}_1 = 0.8$ and $SE(\hat{\beta}_1) = 0.2$. Compute the t-statistic for $H_0 : \beta_1 = 0$.

Worked solution

$$t = 0.8/0.2 = 4.$$

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Statistical significance depends on assumptions about the errors and design.
- A significant slope need not have a large economic magnitude.
- Time-series residual dependence can invalidate elementary standard errors.

Inference for the regression slope — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 If $\hat{\beta}_1 = 0.1$ and $SE = 0.1$, find t .
- 2 What null is tested by a standard slope significance test?
- 3 Why can heteroskedasticity matter for inference?

Inference for the regression slope — solutions

Concise solutions

- ① 1.
- ② $H_0 : \beta_1 = 0$.
- ③ It changes the sampling variance; classical standard errors may be wrong.

Concept

Regression interpretation requires more than a fitted line. Functional form, exogeneity, variance structure and dependence assumptions determine what the coefficients and standard errors mean.

- Omitted variables correlated with included regressors can bias causal interpretations.
- Heteroskedasticity affects conventional variance formulas.
- Serial correlation is common in time-series residuals.
- Prediction, explanation and causal identification are different goals.

Regression assumptions, omitted variables and time-series warnings — example

Problem

A regression of asset return on market return leaves residuals that cluster in high- and low-variance periods. Which assumption is questionable?

Worked solution

Constant conditional variance is questionable; this is a form of heteroskedasticity or volatility clustering.

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not interpret every regression coefficient causally.
- Residual diagnostics are part of model assessment.
- Financial time series often violate independent, identically distributed error assumptions.

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- ① What problem can an omitted confounder create?
- ② What residual feature indicates serial dependence?
- ③ Why is train/test separation useful for forecasting?

Concise solutions

- ① Bias or misleading interpretation of coefficients.
- ② Systematic autocorrelation or runs in residuals.
- ③ It evaluates performance on observations not used to fit the model.

Chapter 15

Bridge to Mathematical Methods for Finance

2 detailed review units $\times$ five frames per unit

From prerequisite objects to course objects — overview

Concept

The course will reinterpret familiar objects in more abstract spaces. The purpose of this package is to make the underlying algebra and calculus automatic before topology, functional analysis, probability structures and time-series methods become more demanding.

- Vectors $\rightarrow$ normed and inner-product spaces.
- Convergence of numbers $\rightarrow$ convergence in metric/function spaces.
- Integrals $\rightarrow$ expectations and continuous probability.
- Linear systems $\rightarrow$ operators, projections and regression.

From prerequisite objects to course objects — example

Problem

Why is understanding the ordinary Euclidean norm useful before studying abstract norms?

Worked solution

It provides a concrete model of distance, size, triangle inequality and convergence. Abstract normed spaces generalize these familiar properties.

From prerequisite objects to course objects — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Do not treat prerequisites as disconnected computations.
- Ask which object is being generalized in every new course topic.
- Return to these pages whenever a new notation hides a familiar operation.

From prerequisite objects to course objects — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- ① Which prerequisite anticipates covariance matrices?
- ② Which prerequisite anticipates expected values of continuous random variables?
- ③ Which prerequisite anticipates Banach-space convergence?

From prerequisite objects to course objects — solutions

Concise solutions

- ① Linear algebra and descriptive/probability covariance.
- ② Definite and improper integration.
- ③ Limits, sequences, completeness and norms.

Final diagnostic checklist — overview

Concept

Before the main course begins, the goal is not to have memorized every page. The goal is to know where each tool lives and to be able to recover it quickly and correctly.

- Can you solve equations, domains, limits, derivatives and nontrivial integrals?
- Can you multiply matrices row-by-column and analyze linear dependence?
- Can you compute and interpret mean, variance, covariance, confidence intervals and p-values?
- Can you distinguish deterministic functions from random variables and sample statistics from parameters?

Final diagnostic checklist — example

Problem

A student cannot remember partial fractions during a probability-density calculation. What should they do?

Worked solution

Return to the integration-techniques section, identify the denominator factorization, decompose the rational function, and then continue the probability calculation. The prerequisite package is designed as a reference, not as a memorization test.

Final diagnostic checklist — checks

Before moving on

These checks are designed to catch the most common conceptual and algebraic errors.

- Mark weak topics before Lecture 1.
- Redo worked examples without looking at the solution.
- Use the exercise solutions only after a serious attempt.

Final diagnostic checklist — exercises

Try these without looking at the next page

The aim is to identify the method before doing the algebra.

- 1 Name three topics you personally need to revise.
- 2 Choose one integral method and state when it is appropriate.
- 3 Explain in one sentence why covariance appears in portfolio variance.

Concise solutions

- ① Individual answer.
- ② Example: integration by parts is appropriate for products where differentiating one factor simplifies it.
- ③ Because variance of a linear combination contains cross-products measuring joint movement.

Prerequisites completed

Next step: Mathematical Methods for Finance

Final recommendation

Do not try to memorize the entire package. Use it as a structured reference: diagnose a gap, return to the relevant five-page unit, reproduce the worked example, and solve the exercises independently.